DEGREE PROJECT

Brand Personality Perception through Sound Symbolism and Semantics

A comparison between Swedish speakers and non-Swedish Speakers

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Abstract

The purpose of this study was to examine how semantic associations and sound symbolism in Swedish product brand names affect the perception of brand personality and compare differences in the perception between Swedish and non-Swedish speakers. The study used a quantitative research design where 24 current product brand names from IKEAs assortment were categorized according to their vowel- and consonant type. The data were collected through two identical questionnaires, one in Swedish and one in English. The data analysis was conducted using an independent t-test and a bivariate correlation test in SPSS. The results indicated that brand personality perceptions differ between Swedish and non-Swedish speakers, which could suggest that language influences how a name is perceived. The results did however only partly support the hypothesis. Semantic associations were statistically significant in three out of four cases while the perception of personality were inconsistent with the hypothesis. The study does however contribute to theory on brand personality and sound symbolism by showing that it can differ across language groups and be more linguistically dependent than previously implied. Practically, the research showed that one cannot solely rely on sound symbolism to convey a certain personality as it can vary between language groups. It can be useful to test the names and see if the desired personality is actually perceived when moving to new markets.

Key words: *Brand personality dimensions, sound symbolism, semantics, brand names*

Sammanfattning

Syftet med denna studie var att undersöka hur semantiska associationer och ljudsymbolik i svenska produktvarumärken påverkar uppfattningen av varumärkespersonlighet, samt att jämföra skillnaden i uppfattning mellan svenska och icke-svenska talare. Studien använde en kvantitativ forskningsdesign där 24 produktnamn från IKEAs aktuella sortiment kategoriserades utifrån vokal- och konsonanttyper. Datan samlades in med two identiska enkäter, en på svenska och en på engelska. Dataanalysen gjordes med ett oberoende t-test samt en bivariat korrelationsanalys i SPSS. Resultaten från dataanalysen indikerade att uppfattningen av varumärkespersonlighet skiljer sig mellan svenska och icke-svenska talare, vilket kan tyda på att språk påverkar hur ett namn uppfattas. Resultaten stödde däremot bara delvik hypoteserna. Semantiska associationer var statistiskt signifikanta i tre av fyra fall medan uppfattningen av varumärkespersonlighet var inkonsistenta med hypotesen. Studien bidrar däremot till forskningen inom varumärkespersonlighet och ljudsymbolik genom att visa att det kan variera mellan språkgrupper och vara mer språkligt påverkade än vad som tidigare antagits. Praktiskt visar studien att ljudsymbolik inte är tillräckligt för att säkerställa en önskad varumärkespersonlighet då det kan variera bland olika språkgrupper. Det kan vara nödvändigt att testa namnen för att se om den önskade personligheten faktiskt uppfattas innan man går in på en ny marknad.

Key words: *Varumärkespersonlighet, ljudsymbolik, semantik, varumärkesnamn*

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1. Introduction

This section will give the background to the study, continuing into the problem discussion, research purpose and questions and ends with the delimitations in the study.

1.1 Background

People are on average exposed to brands and advertisements over 5 000 times every day. You see it in your mail, on your phone, on buildings and just driving on the road (Johnson, 2014). This level of brand exposure has shown that consumers get an emotional meaning tied to brands (Steenkamp, 2017; Zulfugarova, Yüksel, Kurtuluş, & Akturan, 2025). Being a global brand means being recognized by a large majority of people, and it is often these global brands that top the lists of strongest brands (Steenkamp, 2017). Interbrand (2025) listed the top three biggest brands of 2025 to be Apple, Microsoft and Amazon which are brands that are widely recognized by a large majority of the population.

One important aspect of being a recognized brand is the association to it, where the name is one of the first associations people make that creates a positive image for the brand (Klink, 2001). Choosing a brand name in a more recognized language is something that many opt to do to get a more global reach. There are however more factors than the chosen language of the brand name that determine the level of success of a brand name, which in turn has an impact on the probability that a brand will go global. The structure of the name, linguistic attributes and the meaning are some of the factors that affect the quality of the brand name (Zulfugarova, et al., 2025).

A brand name carries meaning and can show some of the features that are offered by the brand. The use of certain letters or sounds can convey a meaning of what a brand is offering (Klink, 2000; Motoki et al., 2022). Klink (2001) claims that brand names that give information in the brand name tend to do better than those who do not. Shrum et al. (2012) found that people prefer when the brand name has information of its attributes in the name when the person was familiar with the particular language of the brand name. The information would however not give meaning to those consumers who do not speak the language the brand name has, which would mean that the name would get lost in translation (Klink, 2000). Klink (2000) on the other hand argues that sound symbolism is not necessarily

tied to language and the meaning could remain even if the consumer does not speak the language. Including information about a product's attributes would therefore create a semantic meaning of a name and would be more linked to the particular language used.

The use of some letters in names show to be associated with smallness and angular shapes or largeness and round attributes which could give the consumer a perception of a brand even if the name is in a language they do not understand (Knoeferle et al., 2017). Aaker (1997) presents the idea that brands are connected to a personality based on attributes of the brands. The five brand personality dimensions that Aaker (1997) presents are; *Sophistication, Sincerity, Excitement, Ruggedness* and *Competence.* As the name of the brand is the first association a consumer has with a particular brand it will be one of the first factors that ties it to a personality. This is important to know when developing a brand name if the goal is to be perceived with a specific personality to attract customers. For example, using vowels that are pronounced in the back of the mouth, like /o/ and /u/, or using consonants that create a harder sound has shown to be more associated with a *rugged* brand personality. On the contrary, using the vowels /i/ and /e/, that are produced at the front of the mouth, or consonants that create a softer sound conveys personalities that are more *sincere* and *sophisticated* (Klink, & Athaide, 2012). The brand personality carries meaning for the brand and sets the brand image (Aaker, 1997). Therefore it is important to set the right intention from the beginning, starting with the name to ensure that the brand will be successful.

1.2 Problem discussion

When developing a brand name, previous research suggests that having an English name will increase the chances of going international (Jang, et al., 2024; Olavarrieta Soto, 2009; Nqobile Kunene, 2025; Malik, 2015). While that can be a shortcut to international success, it is not the only way for a brand to succeed. Interbrand ranked IKEA in 35th place of the best global brands 2025 (Interbrand, 2025). The company is known by a large amount of the population and currently has stores in 62 countries. The name IKEA is an abbreviation of the founders name and place of birth and does not have a direct meaning while still making associations for many people (IKEA, n.d.a). Starting off in a small town in Sweden, IKEA has not changed the way they name their products even with global expansion and still remain successful. While the names of the products are not solely behind the success of the

company, IKEA deviates from previous research's shortcut to success within brand naming. They require all product brand names to have a Scandinavian touch, and preferably the use Swedish letters /å, ä/ and /ö/ in all their product brand names. The names can however not be offensive in any country they are operating in or be associated with politics nor religion (IKEA, n.d.b).

When naming a specific product, IKEA has a system for what kind of name it will have. Couches are for example Swedish locations, bookshelves have male first names and products developed for children are named after animals and nature (IKEA, n.d.b). While this naming system might not be known by Swedish people, the meaning of the names could convey a certain personality dimension because of the consumers' semantic association to the name, e.g. what it actually *means* (Motoki et al., 2022). Non-Swedish speakers on the other hand could be less likely to know that these names are specific locations and would have a perception of the product brand name personality that is based on the phonetics of the name, e.g how it *sounds*. This will be affected by which letters and sounds are in the name, for example the front vowel /i/ is associated with the *Sincerity* and *Sophistication* personality dimensions but also with *smallness* and *roundness* attributes (Klink, & Athaide, 2012; Motoki et al., 2022; Knoeferle et al., 2017). While many market researchers use sound symbolism when naming new brands, there is no evidence that IKEA has such an approach, rather they name their products after actual Scandinavian names and places. That does however not mean that sound symbolism cannot be detected in the names, which leads to the question if phonetics and semantics lead to differences in brand personality perceptions.

Most of the previous research within brand names, brand personality and sound symbolism are done in English, which has a slightly different phonetic structure than Swedish. English has five vowels in its language while Swedish has the same vowels as English, there is the addition of the letter 'Y' which in English is not a vowel as well as three letters that are not in the English alphabet. (Alfabetet, n.d; Majema, n.d). In addition, Swedish has 17 pure vowel sounds while English has 12 (Engstrand, 2007; Saxena, 2017). The difference in phonetics could have an impact on what personality a brand name is perceived between a consumer who speaks Swedish and one who does not. Furthermore, focusing on actual geographic locations can create semantic associations outside of the construction of a name, as individuals who have a connection to a location would have a different perspective than a

person who does not know that the name is a place. A Swedish person could have a semantic connection to a Swedish location, while a non-Swedish speaker would not recognize it as a location.

Previous research has studied the use of sound symbolism in brand naming (Klink, 2000; Klink, 2001; Motoki, et al., 2022). Some studies have connected brand personality and brand names but there is limited research studying the relationship between brand names, sound symbolism and brand personality. The previous research has, to a large extent, been studied in English and while sound symbolism has been studied in the Swedish context, it is limited and has not explored elements of brand personality (Abelin, 2015). Abelin's (2015) study focused on *conventional* sound symbolism which uses clusters of consonants in brand names, but no study has been studied with *synesthetic* sound symbolism which focuses on vowels and consonants. Brand personality dimensions have been used when studying the brand personality among Swedish Universities online branding, but does not regard sound symbolism (Opoku et al., 2008).

As Swedish has almost double the vowels compared to English with the addition of three letters that do not exist in the English alphabet, this differentiation in phonetics may influence the brand perception. While it has been argued that sound symbolism is not only tied to language, semantic perceptions of brand personality could be *dependent* on language while phonetic perception can be *influenced* by language. Furthermore, there is a lack of research on if the semantic perception of a brand personality is equivalent to the phonetic one in a Swedish/non-Swedish context. Previous research has mostly made use of fictitious brand names, but as IKEA is globally known while keeping their Swedish identity the case of IKEA is relevant to investigate as they differ from researchers' understanding of successful global brand naming.

The study can provide insights to future brand naming, as sound symbolism can be integrated to convey a personality in a more global setting without losing the semantic association when a company is trying to express the country-of-origin. It will keep the focus on current IKEA-products with names of geographical locations to see if the brand personality perception differs between Swedish speakers and non-Swedish speakers, as previous research has not clearly shown if brand personality perception changes with sound or meaning.

1.3 Research purpose and hypothesis

The research purpose of the study is to examine how semantic associations and sound symbolism in Swedish product brand names affects the perception of brand personality and compare differences in the perception between Swedish and non-Swedish speakers. To reach the aim two hypothesis were formulated:

H1: *English-speakers will perceive personality based on sound symbolism where names with femininity traits are perceived with the Sophistication and Sincerity dimensions as opposed to the masculinity traits.*

H2: *Swedish-speakers will perceive personality based on semantics if they are familiar with the place.*

1.4 Delimitations

The study is delimited to Swedish product brand names from IKEAs assortment and not the corporate brand name IKEA. This is done because of IKEAs global presence and its use of Swedish in its brand names. The study will not investigate anything further than the names of IKEA and how they are perceived by consumers. It solely focuses on the brand personality perception and does not examine purchase intention, attitudes or buying behaviour of consumers. Only product brand names that are in IKEAs current assortment will be used and the study does not take discontinued products into account. The names will be delimited to products named after a geographical location in Sweden and disregard those names that are located in other countries. Continuing, the research will only regard front- and back vowels and fricative- and stop consonants, central vowels, affricative-, nasal- and approximant consonants will not be examined. Furthermore, the study is limited to two language groups, Swedish speakers and non-Swedish speakers to compare differences in perception between the two.

1.5 Disposition

The organization of the study will go as follows, the next chapter will cover the literature review starting off with brand names and language, semantics and country-of origin within brand naming. It will then discuss sound symbolism and the use of different types of vowels and consonants. Continuing into the five dimensions of brand personalities and the use of sound symbolism and semantics when creating brand names. Then similarities and differences in Swedish and English phonetics will be discussed and the chapter will end in the theoretical framework that will be used.

The next chapter goes over the methodology used in the study, including the research purpose, approach, literature search, data collection, data analysis and ends with the credibility and ethical consideration taken into account when conducting the study. The following chapter includes the data analysis, followed by results.

2. Literature review

This section will present the previous literature and cover brand names, sound symbolism, semantics and brand personality and ends in how all the concepts will work together in the theoretical framework.

2.1 Brand names

When choosing a brand name there are various factors that determine the success, many researches' agree that an effective name is easy to pronounce and is recognizable. But things such as the structure of the name, the language used and the inclusion of product attributes also plays a role in the success of the name (Zulfugova et al, 2025; Abelin, 2015; Cui, 2021). Even if the product has nothing wrong with it, a name that is not meaningful and able to create recognition will not be received well by customers and can lead to failure (Abelin, 2015). As there are thousands of new brand names on the market every year there is a need for companies to spend more and more money on strategies to come up with a name that will ensure success (Argo et al., 2010; Abelin, 2015). The use of certain letters has an effect on consumers ability to recall certain brand names, starting the name with the letters /p, t or k/ which are stop consonants show better name recognition than names starting with fricative consonants as stops are more plosive (Vanden Bergh et al., 1984; Klink, 2000).

2.1.1 Language in brand naming

When choosing the name an effective approach in order to become internationally known is to choose a brand name in a language that is known by a large number of people, like English (Jang et al., 2024). In Latin-America it is popular to choose brand names in English or French to have a global reach (Olavarrieta Soto et al., 2009). This trend is also shown in South Africa where Nguni-owned enterprises chose English names when the goal was to reach people across cultures and borders (Kunene & Mbhele, 2025). Malik (2015) did however conclude that the use of English brand names were preferred over the language that was spoken in the operating country regardless if the goal was to be international or not. The use of the English language in brand names and advertising is in some countries regulated by the government, for example in Iran, China and France. Companies will however favour the use of English, even without the goal to be international, to gain consumer reach within the country

(Talebnejad & Sharifi-Therani, 2025). Specific products can on the other hand be associated with a certain country, for example wine is associated with France while beer is associated with Germany (Yener & Taşçıoğlu, 2021). Therefore a wine with a French name would be more favoured by a consumer than a wine with an English name, even if the consumer is not French, which signals that certain product types are better suited when named in a particular language. This can lead to better memorability of the brand name because the consumer gives more attention to the name than if the name would be in a language they comprehend (Yener & Taşçıoğlu, 2021).

2.1.2 Semantics in brand naming

Semantics in a brand naming context refers to the use of existing words in a brand name to convey meaning, for example using the word "Tech" in part of the brand name for related brands (Klink, 2001).This gives meaning for the brand as the consumer will have an idea what the brand is offering. On the other hand, it can make the brand one of many with similar names and is often time dependent on language (Klink, 2001).

Another way of incorporating semantics is by using size cues in the brand name, like the Mini Cooper which signals to the consumer that the car is on the smaller side. In prior research it has been argued that size can be connected to genders, smallness to femininity and largeness to masculinity. One of the reasons for that is that males are generally larger than females, but it is also shown in languages (Zhang et al., 2022). Speakers of languages that are gendered, like Spanish and German, have certain perceptions of size and gender stereotypes when an object is gendered. For example, the word 'bridge' is feminine in German and masculine in Spanish, in a study the authors found that German speakers described a bridge with attributes connected to femininity (smallness for example) while the Spanish speakers described it with masculine attributes (largeness) which could be influenced by the gender of the word in the respective languages (Boroditsky et al., 2003).

2.1.3 Country of origin

The country of origin can refer to different things, country of *manufacture* (what country is on the made-in label), country of *design* (what country it was designed in) or country of *brand* where the company uses the brand name to signal a country. The country of *origin* is in

the end the country that the consumers associate the brand with disregarding where it was designed and manufactured (Aiello et al., 2009). Companies are known to use the country of origin within their brand when developing their international marketing strategies which in the luxury brands have an impact on consumer decisions (Godey et al., 2012). Oftentimes the country of origin will express a sense of quality, where the perceived quality would be higher for a developed market. Bringing back the example of wine, the perceived quality of wine is higher if the country of origin is France then another country as the product of wine is associated with France (Yener & Taşçıoğlu, 2021).

One way to include the country of origin is to include a geographical location in the name, like Marlboro cigarettes and Virginia Slim cigarettes (Chen et al., 2024). The size of the particular geographic location can signal information of the brand, for example where the brand was founded or the size of the brand. Some bigger companies on the other hand choose to have brand names with smaller geographical locations in the name as it signals authenticity, for example Marlboro cigarettes that are named after a street in London which is a relatively small geographic location compared to the size of the company (Chen et al., 2024). Many of the locations used in IKEAs product brand names are small villages that, even Swedish residents, are completely unaware of. A small village in the north of Sweden with around 600 residents has given the name to one of the most sold book shelves at IKEA, while most consumers owning one would not be able to place it on a map (Broman, 2019; SCB, 2023).

2.2 Sound Symbolism in brand naming

Sound symbolism refers to "the linguistic process in which the sounds of a word provide cues about the word's meaning" (Yorkston & Menon, 2004, p. 43). In other words, a brand name can give information of product attributes based on the phonetic sounds in the name (Motoki et al., 2022). There are different categories in which sound symbolism can create meaning based on different sounds (Klink, 2000). Firstly, there is *Corporeal sound symbolism* which are sounds individuals make to express something physical or emotional, for example saying "auch" when they hurt themselves. The second form is *Imitative* which is used to imitate environmental sounds, for example saying "knock, knock" or different animal sounds like "mjau". *Conventional sound symbolism* is more language dependent than the other categories

but refers to certain clusters of letters that produce words with similar attributes. For example, the cluster “gl” which produces words like: *glitter, glow* and *glimmer* which all share similar attributes. This study will however focus on the last category; *Synesthetic sound symbolism* in which vowels and consonants represent aspects of objects to give meaning to a word (Klink, 2000).

2.2.1 Vowels

In linguistics there are two types of vowels, front and back. The way to deem a vowel front or back is by locating the tongue's highest point when saying a word. If the tongue is in the front of the mouth at its highest point when saying a vowel it is a front vowel and vice versa for back vowels (Klink, 2000). In English the vowels /i/ and /e/ are front vowels, /o/ and /u/ are back vowels while /a/ can be both depending on the length. For example, /a/ is considered a front vowel in the word “hate” but a back vowel in the word “ban” as it produces different sounds (Klink, 2000; Saxena, 2017). Using front vowels often conveys “smallness” while back vowels convey “largeness” (Klink, 2009). This can be perceived because of the degree of which the lips are rounded when produced, back vowels are produced with rounder lips creating a larger opening than front vowels and thus perceived as larger (Elsen et al, 2021).

As front and back vowels are decided by the position of the mouth there is also the categorization of vowels between high or low acoustic frequency vowels, that decides how light the vowel is. High acoustic frequency vowels, such as /i/, can convey attributes of smallness which is shown in many languages by the way they spell the word in that language. “Chiko” in Spanish, “Klein” in German and “Mikros” in Greek are all words that mean smallness where the /i/ is included (Klink, 2000). On the other hand the use of low acoustic frequency vowels, like /o/ conveys largeness. For example, “Gordo” in Spanish, “Groß” in German and “Makros” in Greek which all translate to some kind of ‘largeness’ in English (Klink, 2000). This pattern is also shown in the Swedish language, with the use of /i/ in “liten” and the use of /o/ in “stor” which also translate to “small” and “large”. This shows that sound symbolism is able to travel across languages and create meaning even if a name is not in a language that the consumer can comprehend. The range of acoustic frequency can however vary between languages, whereas it is higher in American English and German and lower in Greek (Jackson & McGowan, 2012).

The majority of previous research in sound symbolism does however not include the high/low acoustic frequency in their research and use front and back vowels instead. As previously stated, the two often overlap. In an experiment it was found that sounds produced with the vowel /i/ (both a front vowel with a high acoustic frequency) were more appropriate for brand names targeting women. On the other hand using the vowel /o/ (a back vowel with low acoustic frequency) was more appropriate when targeting men (Wu et al., 2014). Klink (2009) found in a study that brand names with front vowels were more favored by females than using back vowels. When the same study was made with males, there was no favour between using front and back vowels. His explanation for that was that back vowels are often perceived as less pleasant and unfriendly.

This all together shows that front vowels with high acoustic frequency conveys smallness and femininity along with other attributes that normatively are connected to femininity, such as *lighter*, *milder, thinner, weaker, softer* and *friendlier.* At the same time back vowels with low acoustic frequency conveys largeness and masculinity as well as *darker, heavier* and other attributes often connected to masculinity (Klink, 2000; Klink 2009).

2.2.2 Consonants

When it comes to consonants in sound symbolism they are categorized as stops and fricatives that produce, respectively, hard and soft sounds. Stop consonants include the letters /p, b, t, d, k, g/ and are produced with a closed airflow in the mouth while fricatives, /f, v, s, h, z/ are produced with a small airflow that creates a softer sound as opposed to stops (Motoki et al, 2022). Stops and fricatives can then be further divided into voiced and voiceless stops and fricatives. Voiced refers to the vocal cord vibrating when the sound is produced and vice versa for voiceless (Klink, 2000). This study will however only take stops and fricatives into consideration without the further division. Klink (2000) found that fricatives were perceived as *smaller*, *lighter* and more *feminine* as opposed to stops. Motoki et al (2022) found that stops are linked to brands being evaluated lower than fricatives and were described as *bad, bitter* and *large.* Vanden Bergh (1984) found that brand names starting with a stops were more recognized than brand names starting with a fricative, which could be because of the harder sound that it produces which creates recognition as opposed to fricatives that create a

softer sound. Fricatives are more perceived with the female attributes (*smaller, lighter* and *more feminine)* and have been found to increase the femininity of the overall brand perception of a brand while stops increases the masculine brand personality (Guévremont & Grohmann, 2014).

The main take away from this section is that *femininity* has attributes of *smallness* and *softness* and are perceived when using front vowels and fricatives. *Masculinity* has attributes of *hardness* and *largeness* when brand names contain back vowels and/or stop consonants. This is shown in Table 2.1 below.

Table 2.1 Sound symbolism in brand names

	Femininity	**Masculinity**
Vowel	**Front**	**Back**
Attribute	Smallness	Largeness
Consonante	**Fricative**	**Stop**
Attribute	Softness	Hardness

2.3 Five Dimensions of Brand Personality

Aaker (1997) defines brand personality as "the set of human characteristics associated with a brand." (Aaker, 1997, p. 347). Where the first perception of brand personality for the consumer often starts with the brand name (Cui, 2019). These personality traits for brands are derived from the people that are associated with the brand, which can be the typical user of the brand, the employees and/or the CEO. For example, Virginia Slim cigarettes are perceived as feminine as they are used by women to a larger extent, while Marlboro cigarettes are perceived as manly because of the people who use them the most. Aaker (1997) identified five dimensions of brand personality with different traits corresponding to each dimension. The five are; *Sincerity, Excitement, Competence, Sophistication* and *Ruggedness.*

The dimensions show that there are different things that drive the preference for brands. While the three of the dimensions are derived from human personality dimensions the other two are more connected to what people desire to be, e.g. *sophisticated* or *rugged* (Aaker,

1997). Consumers tend to be drawn to the brand personality themselves have or desire to be (Cui, 2021). The dimensions and its major traits associated are shown in Figure 2.1 below where the first dimension, *Sincerity,* also has traits of honesty, sentimentality and gives an emotion of small-towns. The *Ruggedness* dimension are brands that are perceived as masculine and with toughness. The third dimension, *Competence,* gives the personality of secure, hard working and corporate brands. The *Sophistication* dimension gives traits of femininity, good looking and glamorous. The last dimension *Excitement* is associated with trendy, cool, young and unique brands (Aaker, 1997). The framework developed by Aaker (1997) is generalized from an American perspective, and could be true for American brands. A variation of the framework exists for Chinese personality dimensions, which is a logotypic language where the personality would be perceived from characters (Cui, 2021). This will however not be regarded in this study as Swedish, like English, is a phonetic language although there could be other personalities related to Swedish brand names but as the research is limited for Swedish brands the traits from the original framework will be used.



Figure 2.1 The Five Dimensions of Brand Personality (Aaker, 1997).

2.3.1 Brand personality and sound symbolism

As mentioned in the previous section, front vowels are often related to femininity and smallness while back vowels are related to masculinity and largeness (Klink, 2000; Boroditsky et al., 2003; Knoeferle et al, 2017; Wu et al., 2014; Klink, & Athaide, 2012; Motoki et al., 2022). Connected to the personality dimensions *Ruggedness* and *Sophistication* are masculinity and femininity which also are traits within sound symbolism. Klink and Athaide (2012) concluded that brands with sounds produced with front vowels were more perceived with *Sophistication* personality dimension than back vowels while the *Ruggedness* dimension was perceived with sounds produced with back vowels. They also concluded that brands with the personality *Sincerity* also were produced using front vowels. The study

showed that there is a way to design brand names using sound symbolism to achieve the perception of a particular personality. It did however not explore the use of sound symbolism to the last two personalities, *Competence* and *Excitement.*

Guévremont and Grohmann (2014) tested how consonants (stops vs fricatives) were perceived within brand masculinity and femininity perceptions. Where they related Aakers (1997) dimensions of *Ruggedness* more with stop-constantes because of the masculinity of stops. Fricatives were more related to *Sophistication* and *Sincerity* dimensions. While they used brand masculinity and femininity perceptions there are some overlaps with the traits of Aaker's (1997) dimensions (Masculinity in *Ruggedness,* femininity in *Sophistication* and feminine traits in *Sincerity)* their result only focused on masculinity and femininity and no other traits were identified using consonants. There was therefore no input about the other two brand personality dimensions, *Competence* and *Excitement* in the use of consonants. This could indicate that vowels are more important when creating brand personalities (Kulczynski et al., 2021).

2.3.2 Brand personality and semantics

To give meaning to a brand name for the consumer to have a perception about brand personality it is required that the consumer is able to understand the language (Klink, 2001). Klink (2001) uses semantics in his study for product attribute purposes, like using *Cold* in the brand Coldspot, which signals an attribute of the product for the consumers if they understand the semantic meaning of the word *Cold.* In IKEAs case, the geographic locations they use in their product brand names gives it semantic meaning if the consumer knows the location or can understand that the name refers to a location. To have a perception about brand personality in IKEA products it would be based on phonetic sound symbolism for consumers who do not understand Swedish. For Swedish consumers it would be more based on the perception they have on the actual location, assuming that they know the name is a location. The difference between Swedish and non-Swedish speakers could suggest that there is a difference of brand personality if the perception is based on phonetics or semantics.

2.4 Swedish vs English phonetics

As mentioned, there is a difference in Swedish and English phonetics. Swedish has 9 vowels with 17 pure vowel sounds and English has 5 vowels with 12 vowel sounds. While the English phonetics of front/back vowels and consonants has been discussed the Swedish phonetics also has to be discussed before developing the theoretical framework. Like English, /i, e, short a/ are front vowels and /o, u, long a/ are back vowels but Swedish also has the addition of the extra vowels /y, ä, å, ö/ where /y, ä, ö/ are front vowels and /å/ is a back vowel (Engstrand, 2007). While the same letters are classified the same English vowels are more centralized than Swedish vowels. For example, /o/ in Swedish is produced further back in the mouth than the English /o/, both are however still considered back vowels (Rönnerdal & Johansson, 1992). The difference between a long and a short /a/ can be heard in the words 'bat' and 'father', in 'bat' the a is shorter and produced in the front of the mouth and vice versa for 'father'. In Swedish the short /a/ can be heard in 'hatt' while the long one is heard in 'far' (Rönnerdal & Johansson, 1992).

As mentioned in section *2.2.2 Consonantes*, the previous research in sound symbolism for the most part only takes stops and fricatives into account. There are on the other hand other types of consonants, affricatives are letters that start off being a stop but end in a fricative sound (Engstrand, 2007). Words that start off with affricatives in English are for example '*ch*air', '*sh*are' and '*J*ane'. Affricatives are on the other hand not a part of the Swedish language, which can create pronunciation issues for Swedish speakers learning English (Rönnerdal & Johansson, 1992). Other types of consonants are nasals and approximants, nasals are letters like /m/ and /n/ and approximants are some /r/ and /l/ sounds(Engstrand, 2007). This study will however not take nasals nor approximants into account. Lastly, there are letters that can produce different sounds making them either a stop or a fricative. In Swedish a /c/ can either be pronounced /k/ (stop) in 'camping' or /s/ (fricative) in 'cirka' which also is true for English in 'camping' and 'city'. The letter /x/ on the other hand is only pronounced /ks/ in Swedish while it in English can be pronounced as /ks/ in 'exhibition' or /gz/ before a stressed vowel in 'exhibit'. In both cases they start off as a stop and end in a fricative, it is however not the same thing as affricatives as /ks/ produces two different sounds put together (Rönnerdal & Johansson, 1992).

2.5 Theoretical framework

The study will measure brand personality using sound symbolism and semantics to compare how the brand personality is affected if the respondent speaks Swedish or if has no knowledge of the language. The study will be done in two steps, firstly it will have to identify elements of sound symbolism in Swedish product brand names. For that, the literature review over sound symbolism and Swedish vs English phonetics will be used to categorize the names. Table 2.2 below is an overview of how the categorization will take place.

Table 2.2 Categorization of vowels and consonants between Swedish and English.

	Swedish	**English**
Front vowels	/i, y, e, ö, ä, short a/	/i, e, short a/
Back vowels	/o, u, å, long a/	/o, u , long a/
Stops	/p, t, k, b, d, g, q/	/p, t, k, b, d, g/
Fricatives	/f, v, s, h, tj, sj, w, z/	/f, v, h, s, z, th, sh, zh/
Stops and/or fricatives	/c, x/	/c, j, x/
Disregarding	/j, l, m, n, r/	/l, m, n, q, w, y/

Continuing, using sound symbolism where the previous research shows that brand names containing front vowels and fricatives are associated with *femininity, smallness* and *softness* while back vowels and stops are associated with *masculinity, largeness* and *hardness.* These will be divided into the group *femininity* and the group *masculinity.* Furthermore, the previous research shows that traits from the *femininity* group are represented in the brand personalities *Sincerity* and *Sophistication* while the *masculinity* group is represented in the *Ruggedness* personality dimension. This can be shown with Aakers (1997) example of the Virginia Slims and Marlboro cigarettes where the use of the front vowel /i/ (*feminity)* is used in the Virginia Slims that are more marketed towards women while Marlboro cigarettes uses the back vowel /o/ (*masculinity)* which are perceived as more masculine.

The use of sound symbolism is used to measure the non-Swedish speakers perception of brand personality with Swedish brand names. As non-Swedish speakers would have limited knowledge about the vowels of the language along with the additions of extra letters, an

attempt to pronounce a Swedish name would not regard the Swedish phonetics. The 'rules' of sound symbolism would therefore apply as an attempt to pronounce a Swedish word would be done with the English phonetic structure of vowels and consonantes. The Swedish speakers would on the other hand recognize a word as being in their native language. Therefore the other perspective of measure, semantics, would shape the consumers perception of brand personality based on their association to a certain geographical location they are familiar with. The framework will therefore explain if the rules of sound symbolism apply for non-Swedish speakers when presented with a Swedish word and examines if their perception of brand personality correlates with the Swedish speakers perception or if there are differences between the two.

Figure 2.2 below shows the theoretical framework where semantic associations and sound symbolism lead to a perception of brand personality. It includes the five dimensions of brand personality with the most apparent traits and the two groups of sound symbolism that the study will use along with their respective traits.

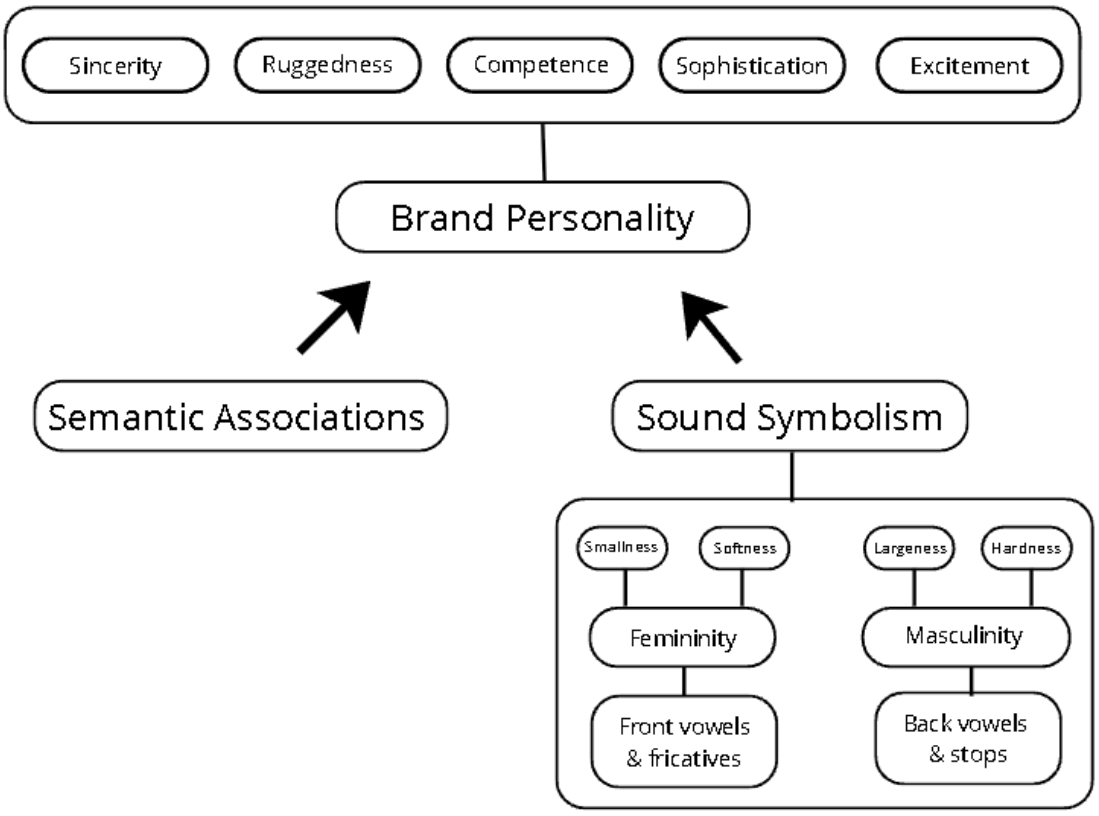


Figure 2.2 Theoretical framework

3. Methodology

The methodology section will explain the research purpose and approach, explain how the literature search was conducted, how the data was collected and sample selected, how the data was analyzed and conclude with the credibility and ethical considerations of the study.

3.1 Research purpose

The research purpose of the study is to examine how semantic associations and sound symbolism in Swedish product brand names affects the perception of brand personality and compare differences in the perception between Swedish and non-Swedish speakers. The study adopts a descriptive research approach by describing what elements of sound symbolism can be found in Swedish brand names. Descriptive research can be used with a quantitative approach to describe results of a phenomenon. It can be conducted with various approaches, two of them being observation and surveys (Purdy, & Popan, 2023). This study used both, observation was used to map out the elements of sound symbolism in Swedish brand names and a survey was conducted to collect data with structured questions. Descriptive research often forms hypotheses derived from previous research which for this study was done after the literature review when finding patterns across different literature (Purdy, & Popan, 2023).

The first hypothesis was derived from research on sound symbolism that the use of front vowels and fricative consonants in brand names were associated with more feminine elements such as; *smallness* and *softness* while the use of back vowels were associated with masculine elements such as; *largeness* and *hardness* (Klink, 2009; Elsen et al, 2021; Wu et al., 2014; Motok et ali, 2022; Guévremont & Grohmann, 2014). Furthermore, previous literature found that front vowels and fricatives, more than back vowels and stop consonants, were connected to the personality dimension *Sophistication* and *Sincerity.* On the other hand, were back vowels and stop consonants, more than front vowels and fricatives, connected to *Ruggedness* (Klink, & Athaide, 2012; Guévremont & Grohmann, 2014). Two groups were then developed; *Femininity* and *Masculinity,* where front vowels and fricatives belonged to the first group and back vowels and stops belonged to the second group. This resulted in the study's first hypothesis:

H1: *English-speakers will perceive personality based on sound symbolism where names with femininity traits are perceived with the Sophistication and Sincerity dimensions as opposed to the masculinity traits.*

The second hypothesis was derived from literature over semantics which refers to the use of existing words in brand names (Klink, 2001). For the consumer to give meaning to the brand name would however require them to know the word beforehand, which makes semantic brand naming language dependent. Using the IKEA brand names can lead to a different personality perception for someone who speaks Swedish than someone who does not. The chosen names are actual Swedish locations and the perception can therefore be based on the location rather than the IKEA brand if the consumer is familiar with the location. The second hypothesis was then formulated:

H2: *Swedish-speakers will perceive personality based on semantics if they are familiar with the place.*

3.2 Research approach

The study was done with a deductive research approach as it was derived from theory to test if the theory supported the outcome (Halvorsen, 1992). Deductive approaches are often combined with qualitative research methodology which has a positivistic viewpoint on research. It assumes that empirical observations are unattached from a theoretical foundation and instead the theoretical foundations act as a base to derive hypotheses from to in turn gain knowledge where the empirical documentation is looked at objectively (Lind, 2019). A quantitative approach enables standardized data to find patterns and connections of a certain phenomena. A qualitative approach would have given multidimensional data which makes it harder to compare data between respondents. As the aim of the study is to compare if perception of brand personality correlates between Swedish and English speakers a quantitative approach is better suited to reach the aim of the study (Lind, 2019).

The downside to a quantitative approach is that it will not get the depth a qualitative study would have of "why" a certain brand personality is perceived for a specific word but instead a

quantitative approach will give the answer of which and give a generalized picture based on the collected data (Halvorsen, 1992). When using a deductive approach it starts off with relevant theory of previous research that is tested with the collected data of the study. The previous literature will be the ground of developing hypotheses, in this case literature over brand personality, sound symbolism and semantics. The hypotheses are then tested via data collection which in this case was done using a survey to see if the hypotheses are supported (Mohammed, 2024).

3.3 Literature search

The literature search was conducted on the data bases provided by the library at Luleå Tekniska Universitet. The main database used were Emerald Journals which has academic journals in Management, Marketing, Economics and Logistics which is suitable for a Master Thesis in International Business. Coursebooks were used for parts of the introduction and methodology section and online-sources such as newspapers and websites were used for the introduction. The online-sources were from recognized newspapers in Sweden along with information from IKEAs own website.

For the literature review the main key words used were: *Brand Personality, Brand Names, Sound Symbolism* and *Semantics* as these are the main concepts of the study. These main concepts were then searched together in different variations such as: “Brand Personality” and “Brand Names” or “Sound Symbolism” and “Brand Personality”. Narrower searches were also conducted with concepts within the theories such as *Synesthetic Sound Symbolism*. Other search words included: *Language, Country of origin, geographic location, vowels, phonetics, Swedish* and *English* among others. With the results of the initial search, a chain search was conducted for the relevant publications to find related articles and further information. The chain search was both from references from the initial searches as well as articles that have cited them.

3.4 Data collection

The data collection was collected in two steps, firstly by a document analysis that guided the second step of a questionnaire. The procedure of both steps will be explained in this section.

3.4.1 Document analysis

Before developing the questionnaire a document analysis was conducted to understand the patterns of sound symbolism. A document analysis entails that the researcher examines data to gain meaning of a phenomena (Bowen, 2009).. The documents can vary, in this case the search was conducted on IKEAs website to gather the names of locations used among their products. The names were selected from different product categories, mainly *couches, armchairs* and *coffee tables.* While many of the products in those categories are named after Swedish locations, all are not. The ones that did not show on a map were therefore excluded, as well as names that were existing locations outside of Sweden, such as *Hemnes* in Norway were also excluded. Repeated names across product categories were treated as one, for example, *Ektorp* is both a couch and an armchair so that name got treated as one. When the selection was completed it ended with a list of 118 names that were analyzed. It was firstly done according to Swedish phonetics, where the vowels of the names were looked at and decided after if there were mostly front vowels or back vowels, the same thing was decided for stops and fricatives. The names were then arranged in four groups, see table 3.1 below:

Table 3.1 Group division

	Front	**Back**
Fricative	Group 1: Femininity	Group 2: Mixed
Stop	Group 3: Mixed	Group 4: Masculinity

The names that had an equal number of either front and back vowels or fricatives and stops were excluded which resulted in 71 names left. The spread of the names is shown in table 3.2 below, where most of the names belonged to the mixed group of front vowels and stop fricatives.

Table 3.2 Spread of names after first sorting

Group 1	*Front & fricative*	Femininity	18
Group 2	*Back & fricative*	Mixed	11
Group 3	*Front & stop*	Mixed	32
Group 4	*Back & stop*	Masculinity	10

The names were then sorted according to English phonetics. Some names could not be deemed to a group based on literature alone, for example as /a/ could be both front and back vowel depending on pronunciation and /ä, å, ö/ which are not in the English alphabet. To gather further understanding about how to deem for example an /a/ as a front or back vowel, a language- and literature professor was consulted along with textbooks on English phonetics.

To understand how a certain name would be pronounced by a non-Swedish speaker, the study analyzed a recording of a non-Swedish speaker going through the list of names. The recording, along with the expertise from the professor and textbooks on English phonetics, contributed to understanding if a name was pronounced using back or front vowels. It also helped to see how different Swedish sound combinations were pronounced with English phonetics. When that analysis was finalized, the names followed the same classification as the Swedish version (see Table 3). This resulted in 52 names left after sorting, the spread after the second sorting is shown below in Table 3.3.

Table 3.3 Spread of names after second sorting.

Group 1	*Front & fricative*	Femininity	11
Group 2	*Back & fricative*	Mixed	6
Group 3	*Front & stop*	Mixed	24
Group 4	*Back & stop*	Masculinity	11

When the second sorting was finalized the group categorization of the Swedish and English phonetic structures were compared. If a name belonged to a different group in the languages they were removed from the study which resulted in 43 names left after the third sorting. The spread is provided in Table 3.4 below.

Table 3.4 Spread of names after third sorting.

Group 1	*Front & fricative*	Femininity	10
Group 2	*Back & fricative*	Mixed	6
Group 3	*Front & stop*	Mixed	21
Group 4	*Back & stop*	Masculinity	6

As there was a gap in the number of names of each group, 4 names were removed from group 1 and 15 were removed from group 4 for each group to have 6 names each. This resulted in 24 names in total, the complete list is shown in alphabetical order in Table 3.5 below.

Table 3.5 The finalized list of names according to their group.

Group 1	**Group 2**	**Group 3**	**Group 4**
Linanäs	Duvholmen	Backsälen	Brobock
Oskarshamn	Hol	Ekerö	Buskbo
Risholmen	Holmsund	Kallax	Ekolsund
Sotenäs	Kungsfors	Långaryd	Gamlehult
Söderhamn	Loshult	Saltsjöbaden	Stockholm
Östavall	Sonhult	Ängesbyn	Tornsborg

3.4.2 Questionnaire

The second step of the data collection was using a questionnaire, in this case the questionnaire was used in a mixed-method study along with the document analysis. A questionnaire is a set of questions that are used to collect data from a population of interest in the particular study (Ranganathan, & Caduff, 2023). Using a questionnaire for this study is positive as it allows to gather larger amounts of data in a shorter amount of time and allows for easier comparison between different groups, in this case between Swedish and non-Swedish speakers (Ranganathan, & Caduff, 2023). The questionnaire in this study was self-administered as the respondents completed the questionnaire without the researcher's assistance. The upside to self-administered questionnaire is that the answers can be more accurate, on the other hand it can lead to the respondent not fully understanding the questions (Ranganathan, & Caduff, 2023).

The questionnaire contained structured questions where the respondents had predecided questions to answer. Using structured questionnaires allows the respondents to provide their answers faster and the researcher to analyze the results easier. On the other hand, there could be other layers of the phenomena that go missing and the respondents are forced to fit their answer although it might not be what they really believe (Ranganathan, & Caduff, 2023). Unlike doing interviews where they can steer in different directions, a questionnaire with

structured questions makes it easier to handle larger amounts of data as they can be coded from words to numbers. It also makes it easier to compare answers between respondents, in this case Swedish speakers and English speakers, as they are presented with the same questions (Halvorsen, 1992).

3.4.2.1 Development of the questionnaire

The questionnaire was developed in two versions, one in Swedish and one in English, according to the aim of the study to be able to compare the two groups studied. The two questionnaires had the same questions but were in different languages to make it as easy as possible for the respondents. Each questionnaire was started with a section of general questions about the respondents and focused mainly on their language abilities. The second section introduced the names listed in Table 7, the names were randomized before added and then further randomized for each participant. The respondent had to answer if they had heard the names before with a 'Yes' or 'No' response. In the last section they then were presented with the traits of each personality dimension from Figure 1 and then choose the personality dimension that they perceived each name to be. As previous research has been limited to three of the five personality dimensions the remaining two were also included to see if there was a pattern connected to the remaining two. The Swedish survey included the same product brand names but assumed that the perception of brand personality was based on semantics if they knew the name before and not on phonetics. The aim was to see if the perceptions aligned if they were based from two different perspectives.

3.4.2.2 Sample selection

The respondents were selected using non-probability sampling which entails that the probability of an individual being part of the sample population is unknown. The downside with using a non-probability sample is that it cannot be said that the sample reflects on the whole unit of analysis (Hagevi & Viscovi, 2016). There are different types of non-probability sampling, the one adopted in this study is quota sampling which has a pre-determined 'quota' of respondents and when the quota is reached no more respondents are allowed to answer (Hagevi & Viscovi, 2016). The unit of analysis for the surveys' are individuals that are either native Swedish speakers or non-Swedish speakers with at most limited knowledge about the Swedish language. The sample population was the quota which consisted of 41 Swedish speakers and 41 non-Swedish speakers.

To reach the quota for the Swedish questionnaire, convenience sampling was performed, which means that the researcher finds the sample that is most easily accessible to them (Golzar et al., 2022). The benefits of using convenience sampling is that it requires less effort to find participants and can be done at a low cost (Golzar et al., 2022). The downside to convenience sampling is that it can create a bias and not be representable to the whole population. To avoid these downsides the researcher can try and control the representativeness of the sample by including diversity or create a larger sample (Golzar et al., 2022). As the research is conducted in Northern Sweden, many of the obtained participants were based in Northern Sweden, which could be seen in the familiarity with the more northern most locations included in the study. To steer away from only having northern based participants, efforts were made to create more diversity.

Firstly, by reaching out to participants of an older age as that showed a greater familiarity with more locations, which could be due to their age. Secondly, snowball sampling was done with the participants of Southern Sweden that initially were acquired using convenience sampling to gather a larger diversity within the country. Snowball sampling refers to acquiring new participants through initial participants to find participants that are harder to reach for the researcher (Ting et al., 2025). Other benefits with this method is that it is cost and time efficient as it relies on prior participants to find new ones and it increases response rates as the participants are familiar and feel comfortable with the research (Ting et al., 2025). The downsides of snowball sampling is that it creates a lack of control over the sample size. But as quota sampling was implemented at the start of the data collection there already was a control over the sample size. Continuing, it could lead to limited representativeness as it leads to participants acquiring new participants with similar characteristics, but as this was the reason to use snowball sampling in the first place that risk was reduced when choosing different initial respondents of different characteristics (Ting et al., 2025).

To reach the quota for the non-Swedish speakers questionnaire, the same combination of convenience and snowball sampling was conducted. This could have helped the risk of lack of diversity as the spread of initial respondents was distributed in a greater part of the world. While the question of native country was not asked, there was indication with the responses

on the question of native language. It showed participants from South America, Europe and Asia along with native English speakers which are spread out across continents.

3.5 Data analysis

To be able to understand the results retrieved from the data collection it is required to categorize the data in order to be able to find patterns and connections (Halvorsen, 1992). As this study is a descriptive and quantitative study a statistical analysis was needed to be able to analyse the data from the survey. This was performed by using two tools, Microsoft Excel and SPSS (Statistical Package for the Social Sciences).

3.5.1 Microsoft Excel

The participants that did not meet the requirements for the questionnaire were removed. This resulted in three participants being removed, one because the terms of the questionnaire were not accepted and the other two because of indicating Swedish as their native language for the non-Swedish speakers version of the questionnaire. When that was completed and all the participants reached the requirements the questionnaires were closed from further answers. They were then transported from Google Forms to Google Sheets which is an automatic feature when using Google Drive programs (Google, n.d). As there were two Google Forms documents, which resulted in two separate Google Sheets documents, the both groups were put into a joint Microsoft Excel Sheet and the coding of the sheet was then begun (See Appendix 1).

For the non-Swedish group their second language was grouped into Germanic languages, such as Norwegian, Danish, German and Dutch with disregard for English that also is a Germanic language but does not contain the extra letters that Norwegian, Danish, German and Dutch do. For those who spoke a Slavic, Latin or Asian language, they were grouped together with the English speakers. To clarify, the non-Swedish participants that spoke a Germanic language besides English were coded to a 1. Those who spoke English and a non-Germanic language were coded to a 2. All the Swedish participants were automatically coded as a 1 despite what their second language was. Further coding included the familiarity with the locations, respondents that were familiar with the location it were coded with a 1,

and if they were not familiar it was coded to a 2. For the personality dimensions they were coded from 1-5 according to this:

1. Sophistication
2. Excitement
3. Sincerity
4. Ruggedness
5. Competence

To be able to see if the personality perceptions based on the groups in Table 3.1 correlated further coding was carried out. An example of respondent 1 is presented in Table 3.6below and shows the coding for group 1. Firstly, a summary of the number of times the respondent answered 'yes' (associated to the familiarity of the location) within a group gave a value between 0-6 as each group contained six locations (Column Grp1 extent knowledge). Zero, meaning that they had never heard of any of the locations and six being that they had heard of all the locations. In the example, respondent 1 was familiar with 4 of the locations. Secondly, a summary of how many times the respondent had chosen each personality within the group was coded to find correlation between the groups in Table 4.2 and the respondents. In the example, P_Linanäs was coded to a 5, meaning that the respondent associated Linanäs with the *Competence*-dimension. In the lower part of the table, Grp1_pers refers to the most associated personality in Group 1, in this case a 3 (Sincerity). The Grp1_Strenght of mode refers to how many times the participant picked *Sincerity,* in this case, three times. This is shown by the upper part in the table where *Sincerity* (3) was chosen for Risholmen, Söderhamn and Östavall. This was carried out for all four groups separately to be able to use it for the tests performed in SPSS. The different groups were separated by numbering, in the example the heading is named with a '1' which changed for every group with a 2, 3 and a 4. Besides coding, Excel was also used to see the distribution of personality perceptions in each group, this is presented in section 4.1.

Table 3.6 Coding

Grp1 extent knowledge	**P_Linanäs**	**P_Oskarshamn**	**P_Risholmen**	**P_Sotenäs**	**P_Söderhamn**	**P_Östavall**
4	5	5	3	2	3	3

Grp1_pers	Grp1_Strength of mode	Sophistication.1	Excitement.1	Sincerity.1	Ruggedness.1	Competence.1
3	3	0	1	3	0	2

3.5.2 Statistical Package for the Social Sciences

The tool SPSS was used for the statistical analysis which is a tool used by students and researchers when having to work with a large amount of data. The statistical tool was chosen due its user-friendly features compared to other tools available. There are four types of variables in SPSS; independent variables, dependent variables, intervening variables and moderator variables (Rahman & Muktatir, 2021). This study does however only have independent variables, the two language groups, and dependent variables, the perception of brand personality dimensions. The dependent variables are influenced by the independent variable, in this case the perception of brand personality is influenced by the language group. But before being able to conduct the tests in SPSS, the data had to be in numerical form for the program to be able to handle the data which was done in the pre-stages in Microsoft Excel. Three types of tests were done using the software, an independent t-test, a bivariate correlation test and a Kruskal-Wallis test.

Firstly, an independent t-test is done when the researcher wants to compare if a dependent variable varies in two different groups (Kent State University, 2026). In this case, Swedish speakers and non-Swedish speakers make up the groups and the independent variable. One requirement to perform the test is that the independent variable has to have exactly two categories and be at a nominal or ordinal level, which these categories do. The dependent variable on the other hand needs to be at an interval or ratio level, the dependent variable here is the number of times the respondent chose one personality which makes it at ratio level. Other requirements are that there can be no missing values, which was avoided by making all the responses mandatory and that the participants cannot not be in both groups, a Swedish person cannot be non-Swedish (Kent State University, 2026). If there is significance it means that there is a difference of brand personality perception between Swedish and non-Swedish speakers.

Secondly, a bivariate correlation analysis was performed to see if there was a correlation between familiarity with a location and brand personality. The analysis is used to examine if there is a relationship between two phenomena (Carpenter, 2017). The Pearson correlation can either show a positive or negative relationship. The further away from zero the value is the more significant the relationship is.

Lastly, a Kruskal-Wallis test was performed to compare if the two language groups had statistically significant differences in brand personality perception. The test is used to compare if two or more groups, in this case two groups, have statistically significant differences on a dependent variable, brand personality (Laerd statistics, n.d). The test requires four assumptions to be met, firstly the dependent variable needs to be at an ordinal or continuous level, in this case the dependent variable is at an ordinal level. The second assumption requires the independent variable to be two or more independent groups, which is met as there are two language groups. The third assumption is that there needs to be an independence of observation, meaning that there cannot be a relationship between the groups and that the groups have the same number of respondents. Each group had N=41 respondents that were not able to participate in both groups, therefore had no relationship. The last assumption is used to know if you can compare median ranks, in this study only the mean ranks were examined however (Laerd statistics, n.d).

3.6 Credibility

To ensure credibility in the study, considerations to validity and reliability have been considered. This section will go over the considerations that have been made to ensure that the different spectrums of validity and reliability have been considered to ensure credibility in the study.

There are four different spectrums to validity in quantitative research, firstly there is the internal validity which considers the relationship between the independent and dependent variables (Sago, n.d). To ensure the internal validity that there could be other factors that affect the perception of brand personality of a certain name, other questions were added such as if the participant had heard of the name before and if they were familiar with the language.

The order in which the names appeared for each respondent was randomized to ensure that the order of the questions did not affect the result.

Secondly, the external validity refers to what degree the results can be generalized to the entire population (Sago, n.d). While it can be hard to gather results to generalize over an entire population, the distribution of the questionnaire was done to ensure that individuals of different ages could respond, as well as individuals that speak different languages. It was not only distributed to one click of individuals for it to get a narrow result in terms of generalizability. While the respondents were met with a personality with some explanation of the traits, they could still vary in the interpretation which could affect the external validity. The sample size can also have an impact on the external validity, for which a sample size of 80 people in total was seen as enough for this study.

Thirdly, when validating if a study actually measures the theories it intends to measure it is called construct validity. In other words, if there is an alignment between the methodology and the theoretical foundation (Sago, n.d). The study intends to measure the perception of brand personality from a semantic perspective and from a perspective of sound symbolism. To ensure the construct validity, expert help was used to ensure the understanding of different phonetic features were correctly understood from a Language and literature professor. To ensure that the phonetic elements could be identified a recording of a non-Swedish speaker was analyzed according to the previous knowledge provided by the professor. Another measure to ensure construct validity was to include the question of if they have heard the names beforehand were included to see if that factor had an impact on their perception. From there it validates if the perception was defined by semantics or sound symbolism. While individuals might have different interpretations of ‘personality’, explanations were provided for the participants to at least have, partly, the same understanding of each personality.

Lastly, to ensure the statistical validity, which refers to how accurate the statistical tests are (Sago, n.d), a sample size of 80 was adopted to ensure variation in the answers. The alternatives were randomized for each participant so that the order would not affect the results and the questions were multiple choice to understand which personality was the most accurate. All the respondents were given the same questions to ensure that the results could be compared in a credible way.

The other consideration to the credibility of the study is to ensure reliability. As with validity, there are different spectrums to reliability as well. Firstly, test-retest validity which refers to if the respondents would have the same perception at different points in time (Sago, n.d). The same questionnaire would be answered by the same respondents to see if the results still align. If they do, the reliability of the study would increase (Sago, n.d). For this study however, a retest was not performed. While that might lower the degree of reliability of the study, in this case there could be many reasons for respondents to change their opinion over time. During the time between the test a participant might visit one of the places and therefore change their opinion of personality, or it could be hard to reach the same individuals as the study adopted anonymity. The participants could also remember their answers from the first time and change their answers based on that. For those reasons, a retest was not performed for this study.

The second spectrum of reliability is inter-rater reliability which means how consistent the phenomena is between individuals that are asked (Sago, n.d). Similar answers ensure a higher reliability while answers that are spread out signals a lower reliability. As the brand personality is measured by two different means it could lead to differences within groups, particularly in the Swedish-group as those answers would be more subjective. For the non-Swedish group, previous research indicates that could is a connection between brand personality and sound symbolism which could indicate more similar answers in this study. If that is proven to be wrong there are different factors that can come into play, for which the control variables have been added to ensure the reliability.

Thirdly, internal consistency reliability shows how different questions measure the same phenomena. It is measured when the researcher wants to examine the same phenomena by using different questions (Sago, n.d). The internal consistency reliability is however not very apparent in this study as there is one question to measure what personality dimension that fits with control variables that could explain the differences in results. The participant is however informed of the different personalities before reaching the questions to ensure that they do not misinterpret the question.

The last spectrum of reliability is parallel forms reliability where two versions of a questionnaire have consistent results when handed out to different participants (Sago, n.d). While this study only has one version of the questionnaire, in two languages, the order of the names are rotated for each participant to ensure that answer fatigue would not be a factor for the later locations in a set list of locations. While that does not ensure parallel forms reliability this type of study would not benefit from the participants receiving different names in different versions, why only one version of the questionnaire was provided. There is however reliability if the participants within the groups have similar answers.

In conclusion, there were considerations of validity and reliability to ensure the credibility of the study. Factors that could impact the results were accounted for, along with considerations for fatigue with randomization of the questions and ensuring a diverse sample. While some parts of reliability are not met, they were not fully adoptable because of the development of the questionnaire.

3.7 Ethical consideration

The study adopted ethical considerations to protect the respondents in the form of them giving their consent to participate in the study and make it clear that their participation was voluntary and they could end the questionnaire before finishing. If they were to be under the age of 18 they could not continue the questionnaire as there are many ethical considerations to take into account when developing questionnaires to children and adolescents. It was therefore decided to not include individuals under 18. The participants were informed about their anonymity when answering the questionnaire in that their answers would not be spread to other sources and would only be used in this study. There was no requirement to enter their names, address or e-mail so that they were ensured that their anonymity would be kept. The general questions were kept at a level so that it would not be traceable to a specific person, there were age groups, instead of specific age, and questions about their language abilities which makes it hard to trace back to one person in case the questionnaire was to be manipulated in any way.

4. Results

The first section of this chapter will show the results of the empirical data that was collected from the questionnaires. A total of 82 responses were included where 41 were native Swedish speakers and 41 were non-Swedish speakers. It will end with an analysis of the results. The chapter will then provide the results of the statistical data analysis where an independent t-test and a bivariate correlation test were performed.

4.1 Empirical data

The questionnaire gathered a total of 82 responses, 41 from Swedish speakers and 41 from non-Swedish speakers. Out of the 82 participants, 51 identified as female and 31 as males. 66 participants were in the 18-29 age group, 6 in the 30-39, 3 in the 40-49 and 7 were in the 50 or older age group. For all the Swedish respondents their native language was Swedish, for the non-Swedish speakers 13 participants listed English as their native language followed by 4 for Spanish. The complete distribution of languages is provided in Table 4.1 below.

Table 4.1 Native language distribution of non-Swedish respondents.

Language	**Number**
English	13
Spanish	4
Chinese	3
German	3
Brazilian Portuguese	2
Dutch	2
Korean	2
Russian	2
Tagalog	2
Greek	1
Hindi	1
Filipino	1

French	1
Latvian	1
Ukrainian	1
Russian	1
Romanian	1

Other languages listed as their second language were Arabic, Tamil, Finnish and Vietnamese. For the Swedish respondents, second languages included English, Spanish and Japanese. 38 of the non-Swedish speakers had no knowledge of Swedish and three had very limited knowledge. Two of them had lived in Sweden for less than four years and the remaining participants had never lived in Sweden.

As the purpose of the study was to examine *how semantic associations and sound symbolism in Swedish product brand names affects the perception of brand personality and compare differences in the perception between Swedish and non-Swedish speakers*, those results will be discussed separately in further detail in the following sections. The chosen locations used in the questionnaire belonged to one out of four groups and the results of the questionnaire will follow the groupings in Table 3.1 below.

Table 4.2 Group division

	Front	**Back**
Fricative	Group 1: Femininity	Group 2: Mixed
Stop	Group 3: Mixed	Group 4: Masculinity

4.1.1 Group 1

The first group, femininity, contained locations that had mostly front vowels and fricative consonants. In the Swedish speaking language group, the most frequently perceived brand personality in Group 1 was *Sincerity* (29,7%), followed by *Ruggedness* (27,6%). *Competence* (29,3%) was the most associated brand personality in group 1 among the non-Swedish participants. The next most was *Sophistication* with 25,6%. In both language groups

Excitement was the least occurring with 10,2% in both groups. The full distribution for group 1 is provided in Figure 4.1 below.

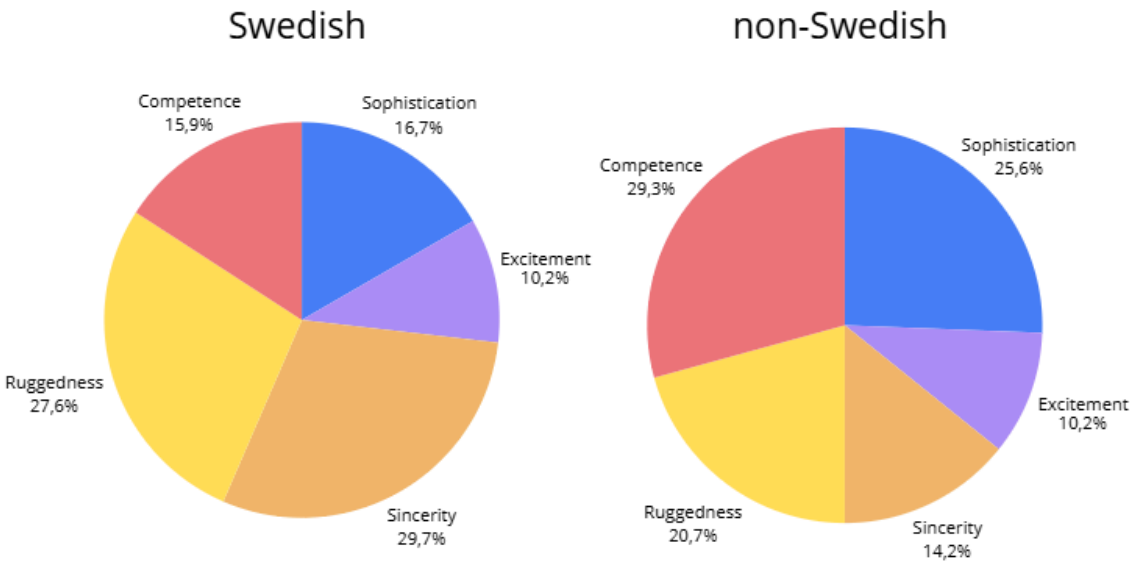


Figure 4.1 Distribution of personality perception in Group 1.

4.1.2 Group 2

Group 2 contained location names that had back vowels and fricative consonants, attributes with elements from both *masculinity* (back vowels) and *femininity* (fricative consonants). As with group 1, the Swedish participants followed the same distribution of perception of brand personality with *Sincerity* (35,3%) and *Ruggedness* (28,0%). The non-Swedish participants continued with *Competence* (30,5%), followed by *Ruggedness* (27,2%). Both language groups remained the lowest with the *Excitement* perception, the Swedish in 10,2% of the cases and the non-Swedish in 9,8% of the cases. The full distribution is provided in Figure 4.2 below.

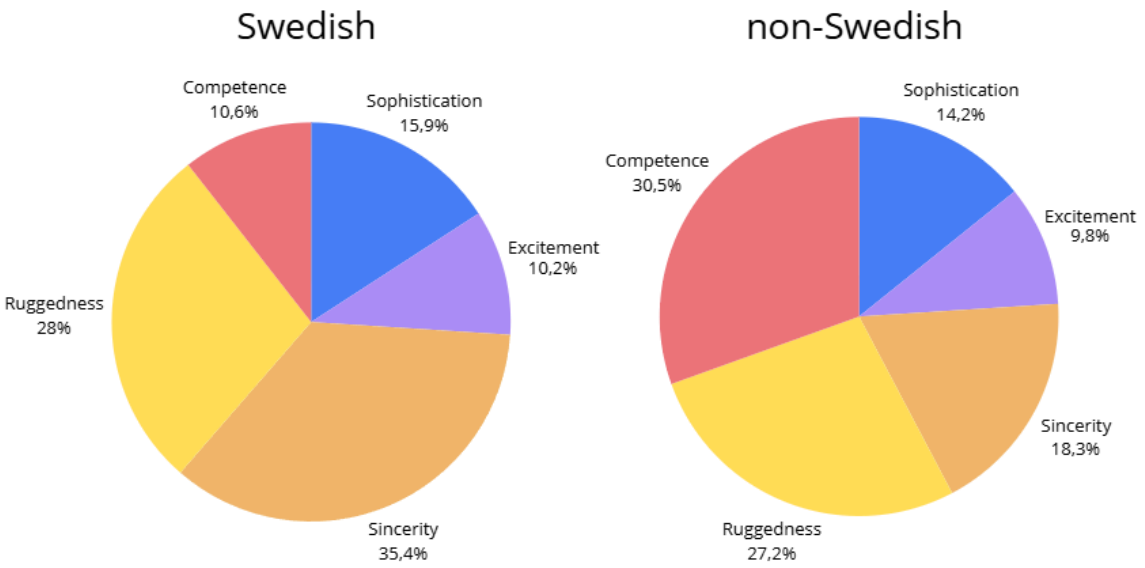


Figure 4.2 Distribution of personality perception in Group 2.

4.1.3 Group 3

The third group also had elements from both *femininity* (front vowels) and *masculinity* (stop consonants). Figure 4.3 shows that the Swedish speakers continued to perceive *Sincerity* (28,5%) as the main brand personality in group 3. This was followed by *Sophistication* (26,8%) while *Excitement* (11%) continued to be the least perceived. For the non-Swedish speakers *Ruggedness* (28,5%) was the most perceived when it came to the combination of front vowels and stop consonants, and was followed by *Competence* (27,2%). *Excitement* continued to be the least perceived for the non-Swedish language group too with 8,9%.

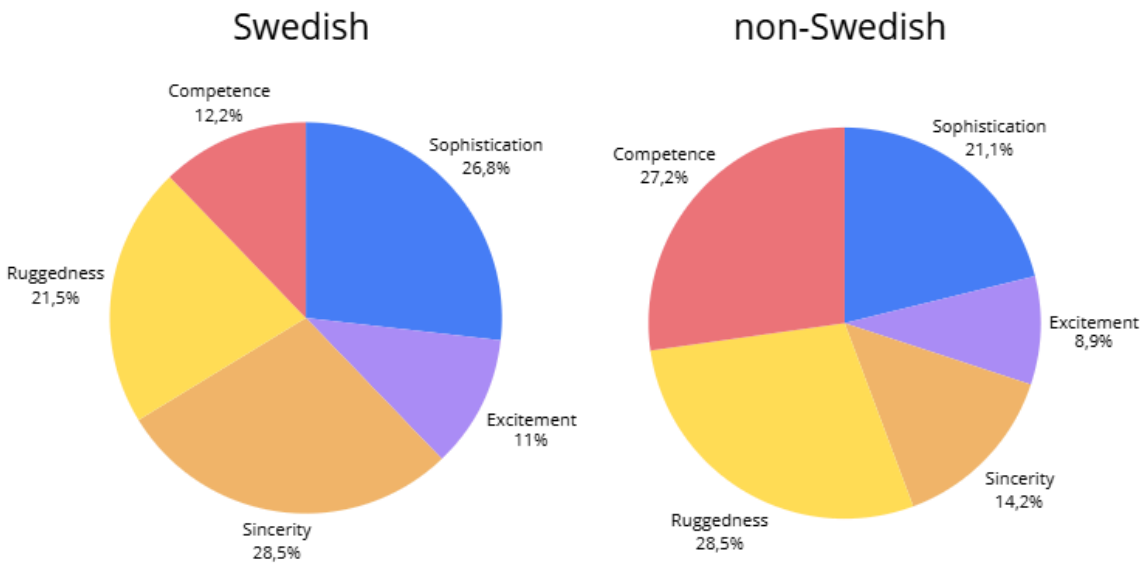


Figure 4.3 Distribution of personality perception in Group 3.

4.1.4 Group 4

The last group, *Masculinity,* had back vowels and stop consonants in the location names. The Swedish speakers perceived those locations to have a *Rugged* (32,1%) brand personality while the non-Swedish language group perceived them to have *Competence* (32,5), that was however followed by *Ruggedness* (27,2%). Both groups still perceived *Excitement* the least. The full distribution for group 4 is in Figure 4.4 below.

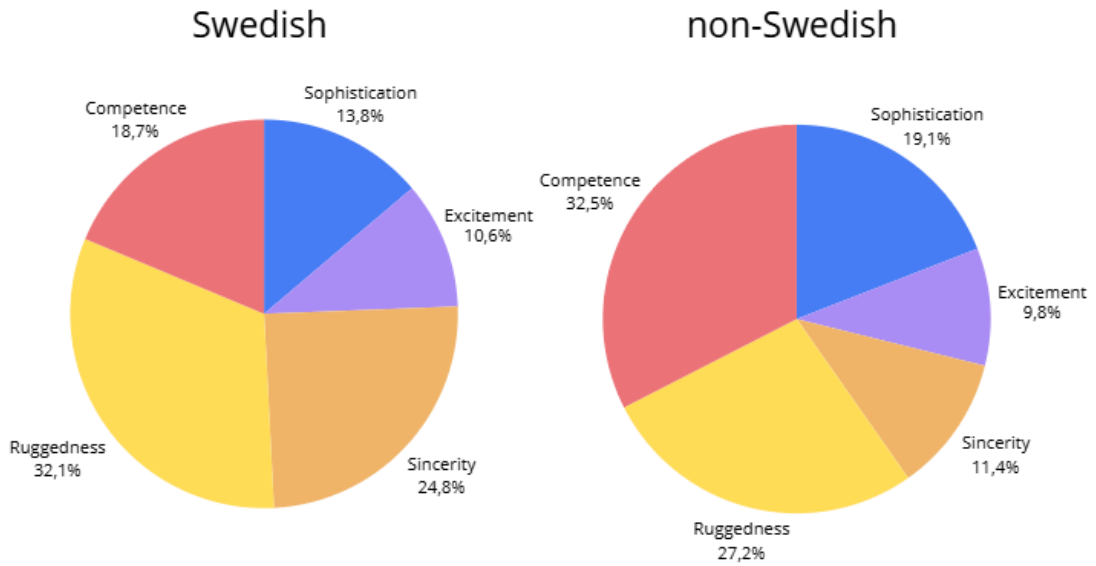


Figure 4.4 Distribution of personality perception in Group 4.

4.1.5 Analysis of the data collection

Previous research has to a majority examined whether the use of front and back vowels are connected to a personality dimension where back vowels were connected to the *Ruggedness* dimension (Klink & Athaide, 2012). The result of the data collection shows that *Ruggedness* is perceived when using back vowels, both in combination with fricative and stop consonants. This was true for both language groups, the non-Swedish group did however in greater extent perceive back vowels in combination with both fricatives and stops with the *Competence* dimension. As that dimension has been excluded in earlier research (Klink & Athaide, 2012; Guévremont & Grohmann, 2014) it could indicate that there could be a connection between back vowels and *Competence.* That dimension did however also seem to be more favourful for the locations with front vowels and fricative consonants that earlier research has pointed to be more associated with *Sophistication* and *Sincerity.* Here, the non-Swedish speakers did put *Sophistication* second most which could correlate with what Klink and Athaide (2012) and Guévremont and Grohmann (2014) found in their studies. For group 3, the non-Swedish speakers found the locations to be most accurate with *Ruggedness* and *Competence* and as those two were most frequent perceptions in three out of four groups it could suggest that the two are more connected when it comes to brand names and sound symbolism than earlier research has suggested.

For the Swedish speaking group, the most frequent brand personality dimensions were *Sincerity* and *Ruggedness.* For the locations with front vowels, the respondent identified them as *Sincere* and *Sophisticated* in combination with fricative and stop consonants respectively. These results are more in line with what earlier research has suggested, that front vowels are more connected to those two dimensions. The dimensions have been found to be more associated with fricatives which for the Swedish respondents remained true as *Sincerity* was found to be the most perceived dimensions among the fricatives. *Ruggedness* has been suggested to be more perceived with stop consonants which was also true for the Swedish speakers. This could suggest that sound symbolism can be apparent across languages, meaning that the same type of sound symbolism found in English can also be true for other languages. The results can however also be influenced by the semantic associations that the

Swedish individuals have to the locations, but the results point to an alignment with earlier research.

To be able to tell if there were significant statistical differences between the language groups an independent t-test was performed and to see if there was a correlation between familiarity with the location and the choice of brand personality, a bivariate correlation was performed. The results of the tests will be presented in the following sections.

4.2 Independent t-test

Four independent t-tests were performed and tested the four different groups separately. The independent t-test was performed using the brand personality dimensions as the dependent variable and the language groups as the independent variable. The test was performed to see if there were significant statistical differences between the two language groups. Table 4.2 below shows that there were significant differences within the groups when it comes to the *Sincerity* and the *Competence* dimensions marked in the green boxes. No statistical significance was found for the remaining dimensions. The full statistical output is provided in Appendix 2. The Swedish results are in line with the previous research as *Sincerity* is connected to front vowels and fricatives, which are in the three first groups. In the last group the dimension was still significant but not as much as in the first three groups, which could suggest that the previous research was correct when it comes to that dimension. For the non-Swedish language group *Competence* was significant in all four groups, although it was higher in group 1. While earlier research does not take this dimension into account, it could imply that the dimension is more associated with the *masculinity* attributes as those are found in back vowels and stop consonants. The only group that did not have those vowels and consonants are group 1, which was the one that had a higher value.

Table 4.2 t- and p-value for all four groups.

	Group 1		Group 2		Group 3		Group 4	
	t-value	**p-value**	**t-value**	**p-value**	**t-value**	**p-value**	**t-value**	**p-value**
Sophistication	−1,810	0,037	0,439	0,331	1,204	0,116	-1,304	0,098
Excitement	0,000	0,500	0,117	0,454	0,593	0,278	0,252	0,401
Sincerity	3,544	< 0,001	3,433	<0,001	3,364	<0,001	3,065	0,001

Ruggedness	1,425	0,079	0,168	0,434	−1,456	0,075	1,046	0,149
Competence	−2,865	0,003	−4,194	<0,001	−3,261	<0,001	−3,198	<0,001

Table 4.3 shows the mean and standard deviation across all four groups further strengthening the differences in the *Sincerity* and *Competence* dimensions. The Swedish group has a higher mean in all four groups in the *Sincerity* dimension than the non-Swedish group and vice versa for the *Competence* dimension, these are marked in green. The biggest differences between the language groups are in group 2 which consists of back vowels and fricatives, where the Swedish respondents were more likely to choose *Sincerity* and the non-Swedish were more likely to choose *Competence.* The results follow the same pattern across all four groups but is the most apparent for group 2.

Table 4.3 Mean-values and standard deviations of t-test.

		Group 1		Group 2		Group 3		Group 4	
	Language group	**Mean**	**Std.dev**	**Mean**	**Std.dev**	**Mean**	**Std.dev**	**Mean**	**Std.dev**
Soph.	Swedish	1	1,36	0,95	1,117	1,61	1,202	0,83	1,046
	non-Swedish	1,54	1,325	0,85	0,882	1,27	1,361	1,15	1,152
Exc.	Swedish	0,61	1,046	0,61	1,093	0,66	1,109	0,63	0,942
	non-Swedish	0,61	0,771	0,59	0,774	0,54	0,711	0,59	0,805
Sinc.	Swedish	1,78	1,423	2,12	1,52	1,71	1,346	1,49	1,468
	non-Swedish	0,85	0,882	1,1	1,158	0,85	0,91	0,68	0,82
Rugg.	Swedish	1,66	1,296	1,68	1,474	1,29	1,101	1,93	1,367
	non-Swedish	1,24	1,338	1,63	1,135	1,71	1,453	1,63	1,157
Comp.	Swedish	0,95	1,182	0,63	0,859	0,73	0,807	1,12	0,98
	non-Swedish	1,76	1,356	1,83	1,515	1,63	1,577	1,95	1,341

4.3 Bivariate correlation test

The bivariate correlation analysis was performed to examine if there was a correlation between familiarity with the location and the personality dimensions with Persons correlation. The values of the test are in Table 4.4 below, where the green marked squares are the values that are statistically significant. For group 1 it shows that greater familiarity resulted in a bigger association with *Sincerity* and a smaller association with *Competence* since that value is negative. It goes with what the other tests have shown, that Swedish

respondents associated group 1 with *Sincerity* more than with *Competence.* In group 2 however, the greater the familiarity the more it was associated with *Sophistication*. It also showed a negative relationship with *Ruggedness* although that was not statistically significant. Since sincerity and *Sophistication* have gone hand in hand in previous research that could be a potential explanation. Here a lower familiarity was not statistically significant with *Competence,* which it is for group 3 along with *Sophistication* again. To summarize, a higher familiarity showed to be significant for the *Sophistication* and *Sincerity* (*femininity)* dimensions in groups 1-3, which also are the three groups that contain the *femininity* attributes. *Competence* was only significant for group 1 and three, which are the two groups that contain front vowels. For the last group, no significant correlation was found. A possible explanation for that is that that is the group that 'Stockholm' belonged in which was the one location that had the most familiarity of all with 75 out of 82 respondents being familiar with it.

Table 4.4 Bivariate correlation analysis.

		Pearson Correlation			
Extent of knowledge	*Sophistication*	*Excitement*	*Sincerity*	*Ruggedness*	*Competence*
Group 1	−0,043	0,034	0,264*	0,001	-0,233*
Group 2	0,274*	0,135	0,006	−0,213	−0,096
Group 3	0,245*	0,082	0,151	−0,209	−0,229*
Group 4	0,134	0,013	−0,039	−0,13	0,044

** Correlation is significant at the 0.05 level*

4.4 Kruskal-Wallis test

The Kruskal-Wallis test was used to see the differences of the ranked personality dimensions between the language groups. The language group was used as the groups and the brand personality dimensions for each combination of front/back and stop/fricative groups as the test fields. The results for group 1 shown in Table 4.5 show that there were statistically significant differences in the dimensions of *Sophistication, Sincerity* and *Competence* where non-Swedish speakers had a higher mean rank for *Sophistication* and *Competence,* while Swedish speakers had a higher mean rank for *Sincerity.* There was not a statistical difference

found for the dimensions *Excitement* and *Ruggedness*. The results show a higher H-value in the dimensions *Sincerity* and *Competence* than in the *Sophistication*-dimensions, meaning that the differences are bigger in those two groups than the latter. In other words, Swedish speakers found the names with front vowels and fricatives to be more *Sincere* while non-Swedish speakers found them *Sophisticated* and especially *Competent.*

Table 4.5 Kruskal-Wallis test, group 1

Dimension	**Mean rank - Swedish**	**Mean rank - Non-Swedish**	**H-value**	**p-value**
Sophistication	35,95	47,05	4,843	0,028
Excitement	40,04	42,96	0,399	0,528
Sincerity	49,50	33,50	9,966	0,002
Ruggedness	45,71	37,29	2,724	0,099
Competence	33,39	49,61	10,320	0,001

The remaining three groups show similar statistically significant differences in terms of personalities. Shown below in Table 4.6, Table 4.7 and 4.8 there are statistical significant differences in the *Sincerity* and *Competence* dimensions. For all three groups, Swedish speakers have ranked *Sincerity* higher than non-Swedish and vice versa for the *Competence* dimension. These results are also apparent for group 1 presented earlier. The biggest difference is found in the *Competence* dimensions for group 2 with a H=17,671 and P=<0,001 which show the biggest separation between the two language groups where Swedish speakers had a mean rank of 30,91 while non-Swedish had 52,09. The lowest statistically significant difference is for group 1 in the *Sophistication* dimension with H=4,843 and P=0,028, followed by the *Competence* dimension for groups 3 and four with H=7,973 and P=0,005.

Table 4.6 Kruskal-Wallis test, group 2

Dimension	**Mean rank - Swedish**	**Mean rank - Non-Swedish**	**H-value**	**p-value**
Sophistication	41,98	41,02	0,037	0,847

Excitement	40,72	42,28	0,114	0,736
Sincerity	49,38	33,62	9,589	0,002
Ruggedness	41,11	41,89	,023	0,879
Competence	30,91	52,09	17,671	<0,001

Table 4.7 Kruskal-Wallis test, group 3

Dimension	**Mean rank - Swedish**	**Mean rank - Non-Swedish**	**H-value**	**p-value**
Sophistication	45,68	37,32	2,721	0,099
Excitement	41,77	41,23	0,013	0,908
Sincerity	49,52	33,48	10,494	0,001
Ruggedness	38,56	44,44	1,325	0,25
Competence	34,41	48,59	7,973	0,005

Table 4.8 Kruskal-Wallis test, group 4

Dimension	**Mean rank - Swedish**	**Mean rank - Non-Swedish**	**H-value**	**p-value**
Sophistication	45,68	37,32	2,721	0,099
Excitement	41,77	41,23	0,013	0,908
Sincerity	49,11	33,89	9,387	0,002
Ruggedness	38,77	44,23	1,145	0,285
Competence	34,41	48,59	7,973	0,005

4.4 Analysis

The aim of the study was to *examine how semantic associations and sound symbolism in Swedish product brand names affects the perception of brand personality and compare differences in the perception between Swedish and non-Swedish speakers*. The following hypothesis were formulated:

H1: *English-speakers will perceive personality based on sound symbolism where names with femininity traits are perceived with the Sophistication and Sincerity dimensions as opposed to the masculinity traits.*

H2: *Swedish-speakers will perceive personality based on semantics if they are familiar with the place.*

To see if the two hypotheses were supported and to reach the aim of the study three statistical tests using SPSS were performed. An independent t-test and a Kruskal-Wallis test were performed to examine if there were statistically significant differences between the two language groups with brand personality perception. The third test was a bivariate correlation test to see if familiarity with the location had a significance with the brand personality perception.

Regarding H1, that seemed to not be the case and the hypothesis was not supported. On the other hand the study showed that *Competence* to a greater extent was connected to the *Masculinity* traits, i.e. back vowels and stop consonants. While *Ruggedness* was the next most chosen personality within the *Masculinity* traits, it was not statistically significant in the t-test and the hypothesis cannot be supported. This is partly inconsistent with what the previous research within sound symbolism and brand personality show. While *Ruggedness* is connected to back vowels and stop consonants as the research suggests (Klink & Athlaide, 2012; Guévremont & Grohmann, 2014), this study in turn suggests that *Competence* is *more* associated with the same dimensions. Their studies did however not regard the use of that dimension although it was mentioned. This study is however based on Swedish names which could give an explanation to the conflicting results. It can suggest that Swedish is seen as more *Competent* to a person who does not understand it.

Regarding H2, it showed that certain dimensions were statistically significant when the respondent was more familiar with the location. As it was based on semantic associations, there was no right or wrong answer based on the theoretical foundation. On the other hand there seemed to be a resemblance among the respondents about their perception which could indicate that more people have the same perception about a location. Familiarity was however not significant in the fourth group for which H2 is only partly supported. The study did

however aim to examine if there were differences between Swedish and non-Swedish speakers' perception of brand personalities, which there was according to the statistical tests. The Kruskal-Wallis test results showed that Swedish speakers to a larger extent perceived the names across all four groups to be more *Sincere* while the non-Swedish speakers found the same names across all four groups to be more *Competent.* In general, the Swedish respondents were more consistent with the previous research in sound symbolism and brand personalities, that could suggest that for a brand name to be successful and be perceived with the wanted personality, the consumer needs to be able to understand the language. This in turn could imply that the 'rules' of sound symbolism are consistent across languages as long as the one to perceive them understands the language.

The only brand personality dimension that was the most overlooked was *Excitement,* as the brand names in the study were geographic locations that could be an explanation as to why. It could be implied that an unknown location is hard to deem as *exciting* when a person has no previous knowledge about it. It could also suggest that there is not any combination of front/back vowels and stops/fricatives consonants that are specifically connected to that dimension.

5. Discussion

This chapter will present the theoretical- and managerial contributions of the research. It will continue with the limitations of the study and end in suggestions for future research.

5.1 Theoretical Contributions

The study contributes to the theory by viewing existing theories of brand personality dimensions and sound symbolism with Swedish brand names. While the hypotheses were not supported by the tests the research adds to existing literature by showing how other dimensions, that previously have been overlooked, are apparent for a person who does not speak Swedish. While *Sincerity* was a dimension that could be linked to traits of femininity, such as front vowels and fricatives (Klink, 2000; Boroditsky et al., 2003), it was also found for other combinations of vowels and consonants. While there were some differences between the different groupings of locations, the overall results showed that the language had a bigger impact on the brand personality perception than the use of different combinations of vowels and consonants. However, it did show that sound symbolism is apparent when the person is able to understand the language, and the choices made by those respondents are consistent with what previous research has implied.

That sound symbolism is able to travel across languages (Klink, 2001) does not seem to be supported in this study as the two language groups had significantly different perceptions in regards to some of the brand personality dimensions. This could have been affected by the brand names being actual geographical locations and that semantic associations would create different perceptions rather than actual phonetic structure (Klink, 2001). The results from this study showed that familiarity with a location was statistically significant in groups 1-3 that included the *feminine* attributes from the theoretical framework. The dimensions chosen here were assumed to be *Sincerity* and *Sophistication* and were those who had a significance. These results may suggest that sound symbolism could have influenced those dimensions as they are within the results of other previous research (Wu et al., 2014; Klink & Athaide, 2012; Guévremont & Grohmann, 2014). Again, it does not however seem to support Klink's (2001) assumption that it can travel across languages when the consumer does not comprehend the language. The results from this study might however be more dependent on

the perception a non-Swedish consumer would get from the actual Swedish language rather than the phonetic structures of the name or location.

Familiarity with the locations, or semantic associations, seemed to bring a similar perception among the respondents. Those who were familiar leaned more to *Sophisticated* and *Sincerity* while unfamiliar leaned more towards *Competence. Competence* was also shown across the language groups to be more perceived in the names with back vowels and stop consonants, *masculinity,* where previous research has assumed *Ruggedness. Compentence* has been overlooked and not directly linked to a certain combination of vowels and consonants to the same degree as *Sophistication, Sincerity and Ruggedness* (Guévremont & Grohmann, 2014; Klink, 2000; Boroditsky et al., 2003; Knoeferle et al, 2017; Wu et al., 2014; Klink, & Athaide, 2012; Motoki et al., 2022). This finding could add to the theory that there could be a connection between *Competence* and *Ruggedness* in the same way that there has been a link found between *Sophistication* and *Sincerity.*

The difference in familiarity and perception of brand personality could also be due to the language groups and that there is a possibility that the Americanized version of the brand personalities that has been criticized before are perceived differently by different language groups (Cui, 2021). Different cultures can have different perceptions of what *Sophisticated* actually means which can be an explanation as to why the perception differed between the two language groups used in this study. This may suggest that an extension of the framework beyond what Cui (2021) did for the Chinese personality dimensions might be needed to examine this field more extensively. This study however presents a Swedish perspective to the brand personality dimensions that are to a degree related to the existing American brand personality dimensions.

5.2 Managerial Contributions

The main managerial contribution is that brand names are perceived differently across language groups and that it can be dependent on language. It is therefore of interest to consider that a brand name that works in the operating country might not be perceived the same way when an organization chooses to take the brand international. As more and more managers choose to incorporate sound symbolism in their brand naming it is worth noting

that it might not travel across languages in the same way as it has been stated before. Before choosing to go international the names should be tested beforehand to ensure that it will be perceptive the way it is intended two as two language groups can have vastly different perceptions of the same name. If the goal is to be perceived with a certain personality, adaptation to the local market might be more flavorful in order to be successful if the same personality is perceived with different attributes in two different locations. The study shows that brand personality perception cannot be predicted using sound symbolism but that it has to be tested using different language groups.

5.3 Limitations

While the study shows that brand personality perception varies across language groups and that sound symbolism is not necessarily able to travel across languages, it still holds some limitations to it.

Firstly, it examined people who speak Swedish with people who do not speak Swedish. As Sweden is a relatively small country compared to the rest of the world it does not fully capture the complexity of languages in general. As the non-Swedish speakers came from a variety of different backgrounds and language abilities, it is not safe to say that the transferability is completely supported across different groups.

Secondly, the sample size was relatively skewed to a younger population, meaning that the generalizability of the results cannot be translated to older age groups. Older people might have had a bigger familiarity with the location, or have created perception about a location over a longer period of time. They could also have had more time to be exposed to more languages which also could have resulted in a different outcome.

The sampling was done using convenience and snowball sampling, while snowball sampling was used to create a bigger diversity the sample could still be concentrated to Northern Sweden as there is a lack of control with that sampling method. This could lead to the familiarity with the more Northern locations being increased while the Southern locations were decreased. The sample size was also rather small and a bigger sample population could have changed the results although there was some significance across the two groups.

Lastly, the locations used in the study varied in size and population which could have affected the familiarity with the locations. It used 'Stockholm', the Swedish capital, that can be assumed to be familiar to people that live outside of Sweden but it also used smaller villages that even for Swedish residents are unknown. Using a more even-sized location could have given the study a more even result even if all the locations used are in the IKEA-assortment. It cannot be assumed that one person will know that.

5.4 Suggestions for Future Research

There are a few suggestions for future research, firstly, comparing native language groups could show how the perception of brand personality might differ between language groups. This could be done with brand names that are not tied to those languages in order to see what differences there are. This takes semantic associations away and can help explain how the same personality types are different in different languages.

Secondly, using a more diverse sample in terms of age as it could affect the familiarity more than what a closer in age sample would. Or change the study to only examine a certain sample size. For this type of study, comparing two age groups that are native Swedish could be useful as it can show how familiarity with more locations changes based on age.

Thirdly, this study only used fricative and stop consonants, which also is true for other previous research. This exudes a number of other consonant types, which also differs across languages. A study examining more types, such as approximants and nasals, could find new patterns that affect brand personality and sound symbolism.

Fourthly, the study could be done to compare if the personality changes if the brand names are more recognizable compared to smaller brand names. This can be done on different scales, global or regional to see if familiarity still has an impact on the perception.

Lastly, the results of this study may have an influence on the country of origin literature, as there is limited research on the country of origin effect on brand personality perception. The results of this study could have been affected by the country of origin effect as people who

are not Swedish and do not speak Swedish might be influenced by their perception of Sweden as a whole and not just based on the name. The results could suggest that Swedish could be perceived as more *Competent* rather than the other dimensions because the respondent had a predetermined view on the country. As the names were taken from a world wide known organization that might have affected the outcome. A study that would examine brand names from different countries that signal the country of origin might deepen the research field on brand personality perception and understand if predetermined perceptions of a country might affect the results.

6. Conclusions

The purpose of the study was to examine how semantic associations and sound symbolism in Swedish product brand names affects the perception of brand personality and compare differences in the perception between Swedish and non-Swedish speakers. It used sound symbolism and semantic associations to compare differences in brand personality perception if the name was familiar or not. The first hypothesis was that the perception would lean more towards the *Sophistication* and *Sincerity* dimensions if the name had front vowels and fricatives as opposed to back vowels and stops that would be more associated with the *Ruggedness* dimension. The second hypothesis was that the perception of brand personality would be affected with semantics if it was a familiar location.

The information needed was collected by a document analysis where 24 product brand names from IKEAs current assortment were categorized in groups according to front vowels, back vowels, fricative consonants and stop consonants. Two questionnaires were then conducted, the questions were identical with the exception for the language, to see what personality type the respondents associated with each location. When all the data was collected it was coded and put into SPSS where an independent t-test and a bivariate correlation analysis was performed.

The results from the tests showed that brand personality perception differs across language groups and that the perception is statistically significant within the language groups. Swedish speakers perceived the groups more *Sincere* and *Rugged* while the non-Swedish speakers perceived it as more *Competent* and *Rugged.*

Regarding the hypothesis, they were not supported as the non-Swedish group did not perceive *Sincerity* and *Sophistication* with front vowels and fricative consonants. Rather, the results showed that hypothesis was supported more with the Swedish speakers. The second hypothesis was partly supported as it showed that three out of four groups were statistically significant when it came to familiarity and perception of brand personality. Overall, the study shows that brand personality perception can be influenced by sound symbolism and semantics, but that the relationships can differ because of differences in linguistic background.

References

Aaker, J. (1997). Dimensions of Brand Personality. *Journal of Marketing Research.* 34(3), 347-356

Abelin, Å. (2015). Phonaesthemes and sound symbolism in Swedish brand names. *Ampersand.* 2, 19-29

Aiello, G., Donvito, R., Godey, B., Pederzoli, D., Wiedmann, K-P., Hennigs, N., Siebels, A., Chan, P., Tsuchiya, J., Rabino, S., Ivanovna, S. I., Weitz, B., Oh, H., & Singh, R. (2009). An international perspective on luxury brand and country-of-origin effect. *Brand Management.* 16(%), 323-337

Alfabetet. (n.d). *Engelska alfabetet.* Alfabetet: https://alfabetet.se/alfabet/engelska

Argo, J. J., Popa, M., & Smith, M. C. (2010). The Sound of Brands. *American Marketing Association.* 74, 97-109

Boroditsky, L., Schmidt, L. A., & Phillips, W. (2003). *Language in Mind: Advances in the Study of Language and Thought.* (3rd ed.). The MIT Press.

Bowen, G. A. (2009). Document Analysis as a Qualitative Research Method. *Qualitative Research Journal.* 9(2), 27-40

Broman, M. K.(December 18th 2019). *Ikeas 12 populäraste produkter genom alla tider.* Leva & Bo: https://levaochbo.expressen.se/inredning/shopping/ikeas-12-popularaste-produkter-genom-alla-tider/

Carpenter, G. W. (2017). Simple Bivariate Correlation. *The SAGE Encyclopedia of Communication Research Method.*

Chen, J., Wang, L., Pang, S., Zhang, Y., & Jiang, Y. (2025). Small = authentic: the effect of geographic brand name size on perceived product authenticity. *Journal of Marketing Theory and Practice.* 33(4), 715-729

Cui, Y. (2019). The presentation of brand personality in English-Chinese brand name translation. *International Journal of Market Research.* 61(1), 33-49

Elsen, H., Németh, R., & Kovács, L. The sound of size revisited - New insights from a German-Hungarian comparative study on sound symbolism. *Language Sciences.* 85, 1-11

Engstrand, O. (2007). *Fonetik light.* (1st ed). Studentlitteratur.

Godey, B., Pederzoli, D., Aiello, G., Donvito, R., Chan, P., Oh, H., Singh, R., Skorobogatykh, I. I., Tsuchiya, J., & Weitz, B. (2012). Brand and country-of-origin effects on consumers' decision to purchase luxury products. *Journal of Business Research.* 65, 1461-1470

Golzar, J., Noor, S., & Tajik, O. (2022). Sampling Method | Descriptive Research. *International Journal of Education.* 1(2), 72-77

Guévremont, A. & Grohmann, B. (2014). Consonants in brand names influence brand gender perceptions. *European Journal of Marketing.* 49(½), 101-122

Hagevi, M., & Viscovi, D. (2016). *Enkäter.* (1st ed). Studentlitteratur.

Halvorsen, K. (1992). *Samhällsvetenskaplig metod.* (Swedish ed.). Studentlitteratur.

IBM. (n.d). IBM SPSS Statistics. IBM: https://www.ibm.com/products/spss-statistics

IKEA. (n.d.a). *Ett ännu bättre IKEA.* IKEA: https://www.ikea.com/se/sv/this-is-ikea/about-us/ett-aennu-baettre-ikea-pubdec78030/

IKEA. (n.d.b). *Hur kommer IKEA på namn för sina produkter?* IKEA: https://www.ikea.com/se/sv/customer-service/knowledge/articles/6f564c4d-2ccc-46de-b643-545a3948dc79.html

Interbrand. (2025). *Best Global Brands 2025.* Interbrand: https://interbrand.com/best-global-brands/global/

Johnson, S. (September 29th 2024). *New Research Sheds Light on Daily Ad Exposures.* SJ Insights: https://sjinsights.net/2014/09/29/new-research-sheds-light-on-daily-ad-exposures/

Kent State University. (May 6th 2026). *SPSS tutorials: Independent Samples T test.* Kent State University: https://libguides.library.kent.edu/spss/independentttest

Klink, R. R. (2000). Creating Brand Names With Meaning: The Use of Sound Symbolism. *Marketing Letters.* 11(1), 5-20

Klink, R. R. (2001). Creating Meaningful New Brand Names: A Study of Semantics and Sound Symbolism. *Journal of Marketing Theory and Practice.* 9(2), 27-34

Klink, R. R. (2009). Gender differences in new brand name response. *Marketing Letters.* 20, 313-326

Klink, R. R., & Athalaide, G. A. (2012). Creating brand personality with brand names. *Marketing Letters.* 23, 109-117

Knoeferle, K., Li, J., Maggioni, E., & Spence, C. (2017). What drives sound symbolism? Different acoustic cues underlie sound-size and sound-shape mapping. *Scientific Reports.* 8(5562),

Kulczynski, A., Brennan, S., & Ilicic, J. (2021). A spokesperson with any name won't be as charming: the phonetic effect of spokesperson name and gender on personality evaluations. *Journal of Brand Management.* 28, 221-239

Laerd Statistics. (n.d). *Kruskal-Wallis H Test using SPSS Statistics.* Retrieved from Laerd Statistics: https://statistics.laerd.com/spss-tutorials/kruskal-wallis-h-test-using-spss-statistics.php

Lind, R. (2019). *Vidga vetandet.* (2nd ed). Studentlitteratur.

Majema!. (n.d). *Vokaler - ordet som betyder sång.* Majema!: https://www.majema.se/blogs/lektioner-svenska/vokaler-ordet-som-betyder-sang?srsltid=AfmBOooS4dZECaGkyOeiQf1VKfGGTe6PyIDJNNkQmnSj2JNR9Zxp1q3G

Malik, S. (2015). Linguistic Iconicity in Brand Names: Derivational Morphology Based Analysis. *Pakistan Journal of Social Sciences.* 35(2), 799-810

Mohammad, S. S. (2024). *Introduction to Research Methods.* International Relations Network.

Motoki, K., Park, J., Pathak, A., & Spence, C. (2022). The connotative meaning of sound symbolism in brand names: A conceptual framework. *Journal of Business Research.* 250, 365-373

Nikolopoulou, K. (June 22nd 2023). *What Is Non-Probability Sampling? Types & Examples.* Scribbr: https://www.scribbr.com/methodology/non-probability-sampling/

Nqobile Kunene, L., & Thokozani Mbhele. (2025). Lost in translation or strategic choice? Unmasking English brand names in Black-Nguni-owned FMCG-SMMEs. *Journal of Research in Marketing and Entrepreneurship*

Olvarietta Soto, S., Manzur Mobarec, E., & Friedmann, R. (2009). Foreign Branding: Examining the Relationship between Language and International Brand Evaluations. *Innovar.* 19(35), 9-18

Opoku, R. A., Hultman, M., & Saheli-Sangari, E.(2008). Positioning in Market Space: The Evaluation of Swedish Universities Online Brand Personalities. *Journal of Marketing for Higher Education. 18(1), 124-14*

Purdy, E, R. & Popan, E. (2023). *Descriptive research.* EBSCO: https://www.ebsco.com/research-starters/social-sciences-and-humanities/descriptive-research#descriptive-research

Ranganathan, P., & Caduff, C. (2023). Designing and validating a research questionnaire - Part 1. *Perspect Clin Res.* 14(3), 152-155

Rönnerdal, G. & Johansson, S. (1992). *Introducing English Pronunciation.* Studentlitteratur.

Sago. (n.d). *The Significance of Validity and Reliability in Quantitative Research.* Sago: https://sago.com/en/resources/blog/the-significance-of-validity-and-reliability-in-quantitative-research/

Saxena, A. (2017). *English Speech Sounds - 12 Pure Vowels and their Symbols.* Explore More and Learn: https://exploremoreandlearn.com/2017/05/english-speech-sounds-12-pure-vowels-and-their-symbols/

Shrum, L.J., Lowrey, T. M., Luna, D., Lerman, D. B., & Liu, M. (2012). Sound symbolism effects across languages: Implications for global brand names. *International Journal of Research in Marketing.* 29(3), 275-279

Statistiska Centralbyrån. (2023). *Tätorter och småorter.* Statistiska Centralbyrån: https://www.scb.se/MI0810

Steenkamp, J-B. (2017). *Global Brand Strategy.* (1st ed.) Springer Nature.

Talebnejad, A., & Sharifi-Tehrani, M. (2025). The effect of national identity on brand name language preference: evidence of nonlinear and moderated moderation effects. *Journal of Marketing Theory and Practice.* 33(2), 191-210

Ting, H., Memon, M. A., Thurasamy, R., & C, J-H. (2025). Snowball Sampling: A Review and Guidelines for Survey Research. *Asian Journal of Business Research.* 15(1), 1-15

Wu, L., Klink, R. R., Guo, J. (2014). Creating Gender Brand Personality with Brand Names: The Effects of Phonetic Symbolism. *Journal of Marketing Theory and Practice.* 21(3), 319-329

Yang, H., Jang, J. J., & Jang, D. Y. (2024). How Brand Names Affect Consumer Perceptions: An Experimental Study on the Sound Symbolism of Foreign Brand Names. *Journal of Global Marketing.* 37(2), 119-137

Yener, D, & Taşçıoğlu, M. (2021). Does the Use of Foreign Languages in Different Types of Products Lead to Different Consumer Perception? *Journal of International Consumer Marketing.* 33(4), 386-398

Yorkson, E., & Menon, G. (2004). A Sound Idea: Phonetic Effects of Brand Names on Consumer Judgement. *Journal of Consumer Research.* 31, 43-51

Zaveri, A. (12th of December 2024). Chi-två-test: Förstå och tillämpa detta statistiska verktyg. Mind the Graph: https://mindthegraph.com/blog/sv_se/chi-square-test/

Zhang, K., Li, S., & Ng, S. (2021). Sizes Are Gendered: The Effect of Size Cues in Brand Names on Brand Stereotyping. *Journal of Consumer Research.* 49

Zulfugarova, N., Yüksel, C. A., Kurtuluş, S., & Akturan, U. (2025). Beyond Words: A Cross-Cultural Comparative Study on Brand Naming Strategies and Sound Symbolism.

Appendix 1 - Questionnaire and coding

The questionnaire first had a question if they agreed to the terms of the study. This was then excluded from the coding as no one who did not accept could not continue the questionnaire.

Q1: What is your native language?

1. Swedish
2. Not Swedish

Q2: What is your second language?

1. English and a non-Germanic language.
2. English and a Germanic language.

Q3: What is your age group?

1. 18-29
2. 30-39
3. 40-49
4. Older than 50

Q4: What gender do you identify with?

1. Female
2. Male

Q5: Have you been taught Swedish in school?

1. Yes
2. No

Q6: What is your level of Swedish?

1. Native language
2. Fluent
3. Conversational level
4. Basic level
5. Very limited knowledge

6. No knowledge

Q7: How long have you lived in Sweden?

1. My whole life
2. 10 years or more
3. 5-9 years
4. Less than 4 years
5. I have not lived in Sweden

Q8: Have you heard of this location? [Linanäs, Oskarshamn, Risholmen, Sotenäs, Söderhamn, Östavall, Duvholmen, Hol, Holmsund, Kungsfors, Loshult, Sonhult, Backsälen, Ekerö, Kallax, Långaryd, Saltsjöbaden, Ängesbyn, Brobock, Buskbo, Ekolsund, Gamlehult, Stockholm, Tornsborg] ***Randomized order for every respondent.***

1. Yes
2. No

Q9: Which personality type fits best for each location? [Linanäs, Oskarshamn, Risholmen, Sotenäs, Söderhamn, Östavall, Duvholmen, Hol, Holmsund, Kungsfors, Loshult, Sonhult, Backsälen, Ekerö, Kallax, Långaryd, Saltsjöbaden, Ängesbyn, Brobock, Buskbo, Ekolsund, Gamlehult, Stockholm, Tornsborg] ***Randomized order for every respondent.***

1. Sophistication
2. Excitement
3. Sincerity
4. Ruggedness
5. Competence

Appendix 2 - Complete statistical output

These are the results of the tests that were done in SPSS and the pictures are directly from SPSS.

2.1 Group 1

Group 1 Statistics

	Language group	N	Mean	Std. Deviation	Std. Error Mean
Sophistication.1	Swedish	41	1,00	1,360	,212
	Non-Swedish	41	1,54	1,325	,207
Excitement.1	Swedish	41	,61	1,046	,163
	Non-Swedish	41	,61	,771	,120
Sincerity.1	Swedish	41	1,78	1,423	,222
	Non-Swedish	41	,85	,882	,138
Ruggedness.1	Swedish	41	1,66	1,296	,202
	Non-Swedish	41	1,24	1,338	,209
Competence.1	Swedish	41	,95	1,182	,185
	Non-Swedish	41	1,76	1,356	,212

Group 1 Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance		Mean	Std. Error	95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Difference	Difference	Lower	Upper
Sophistication.1	Equal variances assumed	1,013	,317	-1,810	80	,037	,074	-,537	,297	-1,127	,054
	Equal variances not assumed			-1,810	79,944	,037	,074	-,537	,297	-1,127	,054
Excitement.1	Equal variances assumed	,913	,342	,000	80	,500	1,000	,000	,203	-,404	,404
	Equal variances not assumed			,000	73,546	,500	1,000	,000	,203	-,404	,404
Sincerity.1	Equal variances assumed	5,681	,020	3,544	80	<,001	<,001	,927	,261	,406	1,447
	Equal variances not assumed			3,544	66,778	<,001	<,001	,927	,261	,405	1,449
Ruggedness.1	Equal variances assumed	,006	,938	1,425	80	,079	,158	,415	,291	-,164	,994
	Equal variances not assumed			1,425	79,922	,079	,158	,415	,291	-,164	,994
Competence.1	Equal variances assumed	,381	,539	-2,865	80	,003	,005	-,805	,281	-1,364	-,246
	Equal variances not assumed			-2,865	78,539	,003	,005	-,805	,281	-1,364	-,246

Correlations

		Grp1Extent knowledge	Sophistication.1	Excitement.1	Sincerity.1	Ruggedness.1	Competence.1
Grp1Extent knowledge	Pearson Correlation	1	-,043	,034	,264*	,001	-,233*
	Sig. (2-tailed)		,704	,761	,016	,990	,035
	N	82	82	82	82	82	82
Sophistication.1	Pearson Correlation	-,043	1	-,113	-,337**	-,383**	-,244*
	Sig. (2-tailed)	,704		,310	,002	<,001	,027
	N	82	82	82	82	82	82
Excitement.1	Pearson Correlation	,034	-,113	1	-,148	-,302**	-,129
	Sig. (2-tailed)	,761	,310		,184	,006	,247
	N	82	82	82	82	82	82
Sincerity.1	Pearson Correlation	,264*	-,337**	-,148	1	-,116	-,391**
	Sig. (2-tailed)	,016	,002	,184		,300	<,001
	N	82	82	82	82	82	82
Ruggedness.1	Pearson Correlation	,001	-,383**	-,302**	-,116	1	-,288**
	Sig. (2-tailed)	,990	<,001	,006	,300		,009
	N	82	82	82	82	82	82
Competence.1	Pearson Correlation	-,233*	-,244*	-,129	-,391**	-,288**	1
	Sig. (2-tailed)	,035	,027	,247	<,001	,009	
	N	82	82	82	82	82	82

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Ranks

	Vad är ditt modersmål?	N	Mean Rank
Sophistication.1	1	41	35,95
	2	41	47,05
	Total	82	
Excitement.1	1	41	40,04
	2	41	42,96
	Total	82	
Ruggedness.1	1	41	45,71
	2	41	37,29
	Total	82	
Sincerity.1	1	41	49,50
	2	41	33,50
	Total	82	
Competence.1	1	41	33,39
	2	41	49,61
	Total	82	

Test Statistics[a,b]

	Sophistication.1	Excitement.1	Ruggedness.1	Sincerity.1	Competence.1
Kruskal-Wallis H	4,843	,399	2,724	9,966	10,320
df	1	1	1	1	1
Asymp. Sig.	,028	,528	,099	,002	,001

a. Kruskal Wallis Test

b. Grouping Variable: Vad är ditt modersmål?

2.2 Group 2

Group 2 Statistics

	Language group	N	Mean	Std. Deviation	Std. Error Mean
Sophistication2	Swedish	41	,95	1,117	,174
	Non-Swedish	41	,85	,882	,138
Excitement2	Swedish	41	,61	1,093	,171
	Non-Swedish	41	,59	,774	,121
Sincerity2	Swedish	41	2,12	1,520	,237
	Non-Swedish	41	1,10	1,158	,181
Ruggedness2	Swedish	41	1,68	1,474	,230
	Non-Swedish	41	1,63	1,135	,177
Competence2	Swedish	41	,63	,859	,134
	Non-Swedish	41	1,83	1,515	,237

Group 2 Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
Sophistication2	Equal variances assumed	,002	,961	,439	80	,331	,662	,098	,222	-,345	,540
	Equal variances not assumed			,439	75,921	,331	,662	,098	,222	-,345	,540
Excitement2	Equal variances assumed	,161	,689	,117	80	,454	,907	,024	,209	-,392	,441
	Equal variances not assumed			,117	72,059	,454	,907	,024	,209	-,392	,441
Sincerity2	Equal variances assumed	,731	,395	3,433	80	<,001	<,001	1,024	,298	,431	1,618
	Equal variances not assumed			3,433	74,728	<,001	<,001	1,024	,298	,430	1,619
Ruggedness2	Equal variances assumed	3,601	,061	,168	80	,434	,867	,049	,290	-,529	,627
	Equal variances not assumed			,168	75,096	,434	,867	,049	,290	-,530	,627
Competence2	Equal variances assumed	5,206	,025	-4,394	80	<,001	<,001	-1,195	,272	-1,736	-,654
	Equal variances not assumed			-4,394	63,309	<,001	<,001	-1,195	,272	-1,739	-,652

Correlations

		Grp2Extent of knowledge	Sophistication 2	Excitement2	Sincerity2	Ruggedness2	Competence2
Grp2Extent of knowledge	Pearson Correlation	1	,274*	,135	,006	-,213	-,096
	Sig. (2-tailed)		,013	,227	,956	,055	,390
	N	82	82	82	82	82	82
Sophistication2	Pearson Correlation	,274*	1	-,042	-,164	-,346**	-,200
	Sig. (2-tailed)	,013		,707	,141	,001	,071
	N	82	82	82	82	82	82
Excitement2	Pearson Correlation	,135	-,042	1	-,300**	-,223*	-,128
	Sig. (2-tailed)	,227	,707		,006	,044	,250
	N	82	82	82	82	82	82
Sincerity2	Pearson Correlation	,006	-,164	-,300**	1	-,262*	-,476**
	Sig. (2-tailed)	,956	,141	,006		,017	<,001
	N	82	82	82	82	82	82
Ruggedness2	Pearson Correlation	-,213	-,346**	-,223*	-,262*	1	-,274*
	Sig. (2-tailed)	,055	,001	,044	,017		,013
	N	82	82	82	82	82	82
Competence2	Pearson Correlation	-,096	-,200	-,128	-,476**	-,274*	1
	Sig. (2-tailed)	,390	,071	,250	<,001	,013	
	N	82	82	82	82	82	82

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Ranks

	Vad är ditt modersmål?	N	Mean Rank
Sophistication2	1	41	41,98
	2	41	41,02
	Total	82	
Excitement2	1	41	40,72
	2	41	42,28
	Total	82	
Sincerity2	1	41	49,38
	2	41	33,62
	Total	82	
Ruggedness2	1	41	41,11
	2	41	41,89
	Total	82	
Competence2	1	41	30,91
	2	41	52,09
	Total	82	

Test Statistics[a,b]

	Sophistication 2	Excitement2	Sincerity2	Ruggedness2	Competence2
Kruskal-Wallis H	,037	,114	9,589	,023	17,671
df	1	1	1	1	1
Asymp. Sig.	,847	,736	,002	,879	<,001

a. Kruskal Wallis Test

b. Grouping Variable: Vad är ditt modersmål?

2.3 Group 3

Group 3 Statistics

	Language group	N	Mean	Std. Deviation	Std. Error Mean
Sophistication3	Swedish	41	1,61	1,202	,188
	Non-Swedish	41	1,27	1,361	,212
Excitement3	Swedish	41	,66	1,109	,173
	Non-Swedish	41	,54	,711	,111
Sincerity3	Swedish	41	1,71	1,346	,210
	Non-Swedish	41	,85	,910	,142
Ruggedness3	Swedish	41	1,29	1,101	,172
	Non-Swedish	41	1,71	1,453	,227
Competence3	Swedish	41	,73	,807	,126
	Non-Swedish	41	1,63	1,577	,246

Group 3 Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance		Mean	Std. Error	95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Difference	Difference	Lower	Upper
Sophistication3	Equal variances assumed	,880	,351	1,204	80	,116	,232	,341	,283	-,223	,906
	Equal variances not assumed			1,204	78,796	,116	,232	,341	,283	-,223	,906
Excitement3	Equal variances assumed	1,158	,285	,593	80	,278	,555	,122	,206	-,287	,531
	Equal variances not assumed			,593	68,095	,278	,555	,122	,206	-,289	,532
Sincerity3	Equal variances assumed	6,699	,011	3,364	80	<,001	,001	,854	,254	,349	1,359
	Equal variances not assumed			3,364	70,241	<,001	,001	,854	,254	,348	1,360
Ruggedness3	Equal variances assumed	9,280	,003	-1,456	80	,075	,149	-,415	,285	-,981	,152
	Equal variances not assumed			-1,456	74,537	,075	,150	-,415	,285	-,982	,153
Competence3	Equal variances assumed	8,457	,005	-3,261	80	<,001	,002	-,902	,277	-1,453	-,352
	Equal variances not assumed			-3,261	59,598	<,001	,002	-,902	,277	-1,456	-,349

Correlations

		Grp3Extent of Knowledge	Sophistication 3	Excitement3	Sincerity3	Ruggedness3	Competence3
Grp3Extent of Knowledge	Pearson Correlation	1	,245*	,082	,151	-,209	-,229*
	Sig. (2-tailed)		,026	,465	,177	,059	,038
	N	82	82	82	82	82	82
Sophistication3	Pearson Correlation	,245*	1	-,036	-,119	-,458**	-,388**
	Sig. (2-tailed)	,026		,746	,288	<,001	<,001
	N	82	82	82	82	82	82
Excitement3	Pearson Correlation	,082	-,036	1	-,204	-,220*	-,261*
	Sig. (2-tailed)	,465	,746		,065	,047	,018
	N	82	82	82	82	82	82
Sincerity3	Pearson Correlation	,151	-,119	-,204	1	-,300**	-,368**
	Sig. (2-tailed)	,177	,288	,065		,006	<,001
	N	82	82	82	82	82	82
Ruggedness3	Pearson Correlation	-,209	-,458**	-,220*	-,300**	1	-,104
	Sig. (2-tailed)	,059	<,001	,047	,006		,352
	N	82	82	82	82	82	82
Competence3	Pearson Correlation	-,229*	-,388**	-,261*	-,368**	-,104	1
	Sig. (2-tailed)	,038	<,001	,018	<,001	,352	
	N	82	82	82	82	82	82

*. Correlation is significant at the 0.05 level (2-tailed).

**. Correlation is significant at the 0.01 level (2-tailed).

Ranks

	Vad är ditt modersmål?	N	Mean Rank
Sophistication3	1	41	45,68
	2	41	37,32
	Total	82	
Excitement3	1	41	41,77
	2	41	41,23
	Total	82	
Sincerity3	1	41	49,52
	2	41	33,48
	Total	82	
Ruggedness3	1	41	38,56
	2	41	44,44
	Total	82	
Competence3	1	41	34,41
	2	41	48,59
	Total	82	

Test Statistics[a,b]

	Sophistication 3	Excitement3	Sincerity3	Ruggedness3	Competence3
Kruskal-Wallis H	2,721	,013	10,494	1,325	7,973
df	1	1	1	1	1
Asymp. Sig.	,099	,908	,001	,250	,005

a. Kruskal Wallis Test

b. Grouping Variable: Vad är ditt modersmål?

2.4 Group 4

Group 4 Statistics

	Language group	N	Mean	Std. Deviation	Std. Error Mean
Sophistication4	Swedish	41	,83	1,046	,163
	Non-Swedish	41	1,15	1,152	,180
Excitement4	Swedish	41	,63	,942	,147
	Non-Swedish	41	,59	,805	,126
Sincerity4	Swedish	41	1,49	1,468	,229
	Non-Swedish	41	,68	,820	,128
Ruggedness4	Swedish	41	1,93	1,367	,214
	Non-Swedish	41	1,63	1,157	,181
Competence4	Swedish	41	1,12	,980	,153
	Non-Swedish	41	1,95	1,341	,209

Group 4 Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance		Mean	Std. Error	95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Difference	Difference	Lower	Upper
Sophistication4	Equal variances assumed	2,819	,097	-1,304	80	,098	,196	-,317	,243	-,801	,167
	Equal variances not assumed			-1,304	79,268	,098	,196	-,317	,243	-,801	,167
Excitement4	Equal variances assumed	,037	,847	,252	80	,401	,802	,049	,194	-,336	,434
	Equal variances not assumed			,252	78,110	,401	,802	,049	,194	-,337	,434
Sincerity4	Equal variances assumed	7,541	,007	3,065	80	,001	,003	,805	,263	,282	1,328
	Equal variances not assumed			3,065	62,725	,002	,003	,805	,263	,280	1,330
Ruggedness4	Equal variances assumed	,540	,465	1,046	80	,149	,299	,293	,280	-,264	,849
	Equal variances not assumed			1,046	77,860	,149	,299	,293	,280	-,264	,850
Competence4	Equal variances assumed	2,880	,094	-3,198	80	<,001	,002	-,829	,259	-1,345	-,313
	Equal variances not assumed			-3,198	73,238	,001	,002	-,829	,259	-1,346	-,312

Correlations

		Extent of knowledge4	Sophistication 4	Excitement4	Sincerity4	Ruggedness4	Competence4
Extent of knowledge4	Pearson Correlation	1	,134	,013	-,039	-,130	,044
	Sig. (2-tailed)		,233	,907	,731	,247	,695
	N	81	81	81	81	81	81
Sophistication4	Pearson Correlation	,134	1	-,172	-,285**	-,266*	-,211
	Sig. (2-tailed)	,233		,123	,009	,016	,057
	N	81	82	82	82	82	82
Excitement4	Pearson Correlation	,013	-,172	1	-,128	-,269*	-,147
	Sig. (2-tailed)	,907	,123		,253	,015	,189
	N	81	82	82	82	82	82
Sincerity4	Pearson Correlation	-,039	-,285**	-,128	1	-,277*	-,381**
	Sig. (2-tailed)	,731	,009	,253		,012	<,001
	N	81	82	82	82	82	82
Ruggedness4	Pearson Correlation	-,130	-,266*	-,269*	-,277*	1	-,317**
	Sig. (2-tailed)	,247	,016	,015	,012		,004
	N	81	82	82	82	82	82
Competence4	Pearson Correlation	,044	-,211	-,147	-,381**	-,317**	1
	Sig. (2-tailed)	,695	,057	,189	<,001	,004	
	N	81	82	82	82	82	82

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Ranks

	Vad är ditt modersmål?	N	Mean Rank
Sophistication4	1	41	45,68
	2	41	37,32
	Total	82	
Sincerity4	1	41	49,11
	2	41	33,89
	Total	82	
Excitement4	1	41	41,77
	2	41	41,23
	Total	82	
Ruggedness4	1	41	38,77
	2	41	44,23
	Total	82	
Competence4	1	41	34,41
	2	41	48,59
	Total	82	

Test Statistics[a,b]

	Sophistication 4	Sincerity4	Excitement4	Ruggedness4	Competence4
Kruskal-Wallis H	2,721	9,387	,013	1,145	7,973
df	1	1	1	1	1
Asymp. Sig.	,099	,002	,908	,285	,005

a. Kruskal Wallis Test

b. Grouping Variable: Vad är ditt modersmål?